

INTEGRALS

$$\frac{d}{dx}(\log x) = \frac{1}{x}$$

i.e. derivative of $\log x$ w.r.t $x = \frac{1}{x}$

$$\therefore \text{antiderivative of } \frac{1}{x} = \log x \Rightarrow \int \frac{1}{x} dx = \log x$$

$\therefore$ Integration is reverse process of differentiation and hence called antiderivative.

$$\frac{d}{dx}(\log x) = \frac{1}{x} + 0$$

$$\frac{d}{dx}(\log x + 1) = \frac{1}{x} \Rightarrow \int \frac{1}{x} dx = \log x + 1$$

$$\frac{d}{dx}(\log x + 5) = \frac{1}{x} \Rightarrow \int \frac{1}{x} dx = \log x + 5 \text{ etc}$$

$$\therefore \int \frac{1}{x} dx = \log x + c$$

$$\text{i.e. } \int f(x) dx = F(x) + c$$

Symbol of integration
elongated 'S',
which represents
summation

any function
of x

constant of integration
integral value

There are two types of integrals:

i) Indefinite

ii) Definite

$$\text{Definite: } \int_a^b f(x) dx$$

Integration is used to find the areas under lines and curves. Entire area is divided into infinitesimally small regions, area of each region is found and added to get the entire area.

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i.e., the small areas are integrated.

$f(x)$	$\int f(x) dx$	$f(x)$	$\int f(x) \cdot dx$
x^n	$\frac{x^{n+1}}{n+1} + C$	k	$kx + C$
$\frac{1}{x}$	$\log x + C$	$\sqrt{x}$	$\frac{2}{3} x^{\frac{3}{2}} + C$
$\frac{1}{x^2}$	$-\frac{1}{x} + C$	$\frac{1}{\sqrt{x}}$	$2\sqrt{x} + C$
e^x	$e^x + C$	a^x	$\frac{a^x}{\log a} + C$
$\sin x$	$-\cos x + C$	$\cos x$	$\sin x + C$
$\tan x$	$\log(\sec x) + C$	$\cot x$	$\log(\sin x) + C$
$\sec x$	$\log(\sec x + \tan x) + C$	$\operatorname{cosec} x$	$\log(\operatorname{cosec} x - \cot x) + C$
$\sec^2 x$	$\tan x + C$	$\operatorname{cosec}^2 x$	$-\cot x + C$
$\sec x \cdot \tan x$	$\sec x + C$	$\operatorname{cosec} x \cdot \cot x$	$-\operatorname{cosec} x + C$
$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x + C$	$\frac{1}{1+x^2}$	$\tan^{-1} x + C$
$\frac{1}{x\sqrt{x^2-1}}$	$\sec^{-1} x + C$		

- 1) $\int x \cdot dx = \frac{x^2}{2} + C$
- 2) $\int x^7 dx = \frac{x^8}{8} + C$
- 3) $\int x^{100} dx = \frac{x^{101}}{101} + C$
- 4) $\int x^5 dx = \frac{x^6}{6} + C$
- 5) $\int \frac{1}{x} dx = \log x + C$
- 6) $\int \frac{1}{x^7} dx = \int x^{-7} dx = \frac{x^{-6}}{-6} + C$
- 7) $\int \frac{1}{x^5} dx = \int x^{-5} dx = \frac{x^{-4}}{-4} + C$
- 8) $\int \frac{1}{x^{11}} dx = \frac{x^{-10}}{-10} + C$
- 9) $\int \frac{1}{x^{100}} dx = \frac{x^{-99}}{-99} + C$
- 10) $\int \sqrt[3]{x^2} \cdot dx = \int x^{\frac{2}{3}} dx = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} + C$

$$11) \int \frac{1}{\sqrt[3]{x^2}} dx = \int x^{-2/3} dx = \frac{3x^{1/3}}{1} + C$$

$$12) \int \sqrt{x^6} dx = \int x^{3/2} dx = \frac{2x^{5/2}}{5/2} + C$$

$$13) \int \frac{1}{\sqrt{x^5}} dx = \int x^{-5/2} dx = \frac{2x^{-3/2}}{-3/2} + C$$

$$14) \int \frac{1}{\sqrt{x^6}} dx = \int x^{-3} dx = \frac{x^{-2}}{-2} + C$$

$$15) \int \sqrt[3]{x^2} dx = \int x^{2/3} dx = \frac{3x^{5/3}}{5/3} + C$$

$$16) \int \frac{x^2 + 2x}{x} dx = \int x + 2 dx = \frac{x^2}{2} + 2x + C$$

$$17) \int \frac{3x^2 + 2x^5 + 7x}{x^2} dx = \int 3 + 2x^3 + \frac{7}{x} dx = 3x + \frac{2x^4}{4} + 7 \log x + C$$

$$18) \int 5x^3 - 7x^2 + 5x - 5 dx = \frac{5x^4}{4} - \frac{7x^3}{3} + \frac{5x^2}{2} - 5x + C$$

$$19) \int \left(x + \frac{1}{x}\right)^2 dx = \int x^2 + \frac{1}{x^2} + 2 dx = \frac{x^3}{3} - \frac{1}{x} + 2x + C$$

$$20) \int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx = \int x + \frac{1}{x} - 2 dx = \frac{x^2}{2} + \log x - 2x + C$$

$$21) \int (3x - 5)^2 dx = \int 9x^2 - 30x + 25 dx = \frac{9x^3}{3} + \left(-\frac{30x^2}{2}\right) + 25x + C$$

$$22) \int \frac{2x-1}{\sqrt{x}} dx = \int 2\sqrt{x} - \frac{1}{\sqrt{x}} dx = 2\left(\frac{2}{3}\right)x^{3/2} - 2\sqrt{x} + C$$

$$23) \int \frac{(3-x)^2}{\sqrt{x}} dx = \int \frac{9 + x^2 - 6x}{\sqrt{x}} dx = \int \frac{9}{\sqrt{x}} + \sqrt{x} - 6\sqrt{x} dx$$

$$= 9(2\sqrt{x}) + \frac{x^{3/2}}{3/2} - 6\left(\frac{2}{3}\right)x^{3/2} + C$$

$$24) \int x(1-\sqrt{x}) dx = \int x - x^{3/2} dx = \frac{x^2}{2} - \frac{x^{5/2}}{5/2} + C$$

$$25) \int \sqrt{x} (x^2 - 5x + 1) dx = \int x^{9/2} - 5x^{5/2} + \sqrt{x} dx = \frac{2}{7} x^{7/2} - 5 \cdot \frac{2}{5} x^{5/2} + \frac{2}{3} x^{3/2}$$

$$26) \int 3\sqrt{x} (3x^2 - 5x + 7) dx = \int 3(3x^{5/2} - 5x^{3/2} + 7\sqrt{x}) dx$$

$$= 9 \cdot \frac{2}{7} x^{7/2} - 15 \left(\frac{2}{5} \right) x^{5/2} + 21 \left(\frac{2}{3} \right) x^{3/2} + C$$

$$27) \int (1-x)^3 dx = \int (1 - 3x + 3x^2 - x^3) dx = x - \frac{3x^2}{2} + x^3 - \frac{x^4}{4} + C$$

$$28) \int 3\sin x + \tan x - 4\operatorname{cosec} x dx = -3\cos x + \log(\sec x) - 4\log(\operatorname{cosec} x - \cot x)$$

$$29) \int 3\cos x - \cot x + 5\sec x dx = 3\sin x - \log(\sin x) + 5\log(\sec x + \tan x) + C$$

NOTE: * $\int \frac{1}{1 \pm \sin/\cos} \Rightarrow$ rationalise

$$* \int \frac{\sin^2 x}{\cos x} = \int \frac{\sin x}{\cos x} \cdot \sin x$$

$$* \int \frac{\sin x}{\cos^2 x} = \int \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \int \sec x \cdot \tan x$$

$$* \int \frac{\cos x}{\sin^2 x} = \int \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = \int \operatorname{cosec} x \cdot \tan x$$

$$* \frac{1}{\sqrt{a^2 - x^2}} \Big/ \sqrt{a^2 - x^2}, \text{ put } x = a \sin \theta / a \cos \theta$$

$$* \frac{1}{a+x^2} \Big/ \sqrt{a^2 + x^2} \Big/ \frac{1}{\sqrt{a^2 + x^2}}, \text{ put } x = a \tan \theta$$

$$* \frac{1}{\sqrt{x^2 - a^2}} \Big/ \sqrt{x^2 - a^2} \Big/ \frac{1}{x^2 - a^2}, \text{ put } x = a \sec \theta / a \operatorname{cosec} \theta$$

$$* \int \sin^2 \rightarrow \frac{1 - \cos 2x}{2}$$

$$* \int \cos^2 \rightarrow \frac{1 + \cos 2x}{2}$$

$$\begin{aligned} * \sin^3 \theta &= \frac{3 \sin \theta - \sin 3\theta}{4} \\ * \cos^3 \theta &= \frac{3 \cos \theta + \cos 3\theta}{4} \\ * \sin^4 &= (\sin^2)^2 = \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1 + \cos^2 2x - 2 \cos 2x}{4} \\ &= \frac{1}{4} \left[1 + \frac{1 + \cos 4x}{2} - 2 \cos 2x \right] \\ \int \sin^4 &= \int \frac{1}{4} \left[\frac{3}{2} + \frac{\cos 4x}{2} - 2 \cos 2x \right] \\ * \cos^4 &= (\cos^2)^2 = \left(\frac{1 + \cos 2x}{2} \right)^2 + \dots \\ * \tan^2 x &= \sec^2 x - 1 \\ * \cot^2 x &= \operatorname{cosec}^2 x - 1 \\ * \tan^3 &= \tan^2 \cdot \tan = (\sec^2 x - 1) \tan x = \tan x \cdot \sec^2 x - \tan x \dots \\ * \cot^3 &= \cot^2 \cdot \cot = (\operatorname{cosec}^2 x - 1) \cot x = \cot x \cdot \operatorname{cosec}^2 x - \cot x \end{aligned}$$

Integration by Substitution:

$$\begin{aligned} \int \sin x \cdot dx &= -\cos x \\ * \int \sin 2x \cdot dx & \quad 2x = t \\ &= \int \sin t \cdot \frac{dt}{2} \\ &= -\frac{1}{2} \cos t + C \\ &= -\frac{1}{2} \cos 2x + C \\ * \int e^{3x} \cdot dx &= \frac{e^{3x}}{3} + C \\ * \int \tan 2x \cdot dx &= \frac{\log(\sec 2x)}{2} + C \end{aligned}$$

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Eg: $\int (2x+1)(3x^2+3x+7)^5 \cdot dx$ $3x^2+3x+7 = t$ $6x+3 = \frac{dt}{dx}$ $3(2x+1)dx = dt$ $(2x+1)dx = \frac{dt}{3}$

$$I = \int \frac{1}{3} t^5 dt$$

$$= \frac{1}{3} \frac{t^6}{6} + C$$

$$= \frac{1}{18} (3x^2+3x+7)^6 + C$$

Eg: $\int \frac{(\log x)^6}{x} \cdot dx$ $\log x = t$ $\frac{1}{x} \cdot dx = dt$

$$\int t^6 \cdot dt$$

$$= \frac{t^7}{7} + C = \frac{(\log x)^7}{7} + C$$

Eg: $\int (\tan^2 x - 3 \tan x + 5) \sec^2 x \cdot dx$ $\tan x = t$ $\sec^2 x \cdot dx = dt$

$$= \int (t^2 - 3t + 5) \cdot dt$$

$$= \frac{t^3}{3} - 3 \frac{t^2}{2} + 5t + C$$

$$= \frac{\tan^3 x}{3} - \frac{3}{2} \tan^2 x + 5 \tan x + C$$

* $\int [f(x)]^n \cdot f'(x) \cdot dx = \frac{[f(x)]^{n+1}}{n+1} + C$

* $\int \frac{f'(x)}{f(x)} \cdot dx = \log[f(x)] + C$

EXERCISE 7.1

Eg 1] i) $\int \cos 2x \cdot dx = I$

$$I = \frac{\sin 2x}{2} + C$$

Eg 2] ii) $I = \int (3x^2 + 4x^3) dx = \frac{3x^3}{3} + \frac{4x^4}{4} + C = x^3 + x^4 + C$

iii) $I = \int \frac{1}{x} \cdot dx = \log x + C$

$$\text{Eg 2] i) } I = \int \frac{x^3 - 1}{x^2} dx = \int \left(x - \frac{1}{x^2}\right) dx = \frac{x^2}{2} + \frac{1}{x} + C$$

$$\text{iii) } I = \int \left(x^{3/2} + 2e^x - \frac{1}{x}\right) dx = \frac{2}{5}x^{5/2} + 2e^x - \log x + C$$

$$\text{Eg 3] i) } I = \int (\sin x + \cos x) dx = -\cos x + \sin x + C$$

$$\text{ii) ** } I = \int \operatorname{cosec} x \cdot (\operatorname{cosec} x + \cot x) dx = \int (\operatorname{cosec}^2 x + \operatorname{cosec} x \cdot \cot x) dx \\ = -\cot x - \operatorname{cosec} x + C$$

$$\text{18) ** } I = \int \sec x (\sec x + \tan x) dx = \int (\sec^2 x + \sec x \cdot \tan x) dx \\ = \tan x + \sec x + C$$

$$\text{iii) ** } I = \int \frac{1 - \sin x}{\cos^2 x} dx = \int \frac{1}{\cos x} - \frac{\sin x}{\cos x \cdot \cos x} dx \\ = \int (\sec^2 x - \sec x \cdot \tan x) dx = \tan x - \sec x + C$$

$$\text{** } I = \int \frac{1 - \cos x}{\sin^2 x} dx = \int (\operatorname{cosec} x - \operatorname{cosec} x \cdot \cot x) dx \\ = -\cot x + \operatorname{cosec} x + C$$

$$\text{1) } \int \sin 2x = -\frac{\cos 2x}{2} + C$$

$$\text{2) } I = \int \cos 3x = \frac{\sin 3x}{3} + C$$

$$\text{3) } I = \int e^{2x} \cdot dx = \frac{e^{2x}}{2} + C$$

$$\text{4) } I = \int (ax + b)^2 = \int (a^2 x^2 + b^2 + 2abx) dx \\ = \frac{a^2 x^3}{3} + b^2 x + \frac{2abx^2}{2} + C$$

$$\text{5) } I = \int \sin 2x - 4e^{3x} = -\frac{\cos 2x}{2} - \frac{4e^{3x}}{3} + C$$

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$$6) \int (4e^{3x} + 1) dx = \frac{4e^{3x}}{3} + x + C$$

$$7) \int x^2 \left(1 - \frac{1}{x^2}\right) dx = \int (x^2 - 1) dx = \frac{x^3}{3} - x + C$$

$$8) \int (ax^2 + bx + c) dx = \frac{ax^3}{3} + b\frac{x^2}{2} + cx + C$$

$$9) \int (2x^3 + e^x) dx = \frac{2x^4}{4} + e^x + C$$

$$10) \int \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 dx = \int \left(x + \frac{1}{x} - 2\right) dx = \frac{x^2}{2} + \log x - 2x + C$$

$$11) \int \frac{x^3 + 5x^2 - 4}{x^2} dx = \int \left(x + 5 - \frac{4}{x^2}\right) dx$$

$$= \frac{x^2}{2} + 5x + 4\frac{1}{x} + C$$

$$12) \int \frac{(x^3 + 3x + 4)}{\sqrt{x}} dx = \int \left(x^{\frac{5}{2}} + 3\sqrt{x} + \frac{4}{\sqrt{x}}\right) dx$$

$$= \frac{2}{7} x^{\frac{7}{2}} + 3\left(\frac{2}{3}\right) x^{\frac{3}{2}} + 4(2\sqrt{x}) + C$$

$$13) \int \frac{(x^3 - x^2 + x - 1)}{x-1} dx = \int \frac{x^2(x-1) + 1(x-1)}{(x-1)} dx$$

$$= \int (x^2 + 1) dx = \frac{x^3}{3} + x + C$$

$$14) \int (1-x)\sqrt{x} dx = \int (\sqrt{x} - x^{3/2}) dx = \frac{2}{5} x^{5/2} - \frac{2}{7} x^{7/2} + C$$

$$15) \int \sqrt{x} (3x^2 + 2x + 3) dx = \int (3x^{5/2} + 2x^{3/2} + 3\sqrt{x}) dx$$

$$= 3 \cdot \frac{2}{7} x^{7/2} + 2 \cdot \frac{2}{5} x^{5/2} + 3 \cdot \frac{2}{3} x^{3/2} + C$$

$$16) \int (2x - 3\cos x + e^x) dx = x^2 - 3\sin x + e^x + C$$

$$17) I = \int (2x^2 - 3\sin x + 5\sqrt{x}) dx = \frac{2x^3}{3} + 3\cos x + 5 \cdot \frac{2}{3} x^{3/2} + C$$

$$18) I = \int \sec x (\sec x + \tan x) dx = \int (\sec^2 x + \sec x \tan x) dx \\ = \tan x + \sec x + C$$

$$19) I = \int \frac{\sec^2 x}{\csc^2 x} dx = \int \frac{\sin^2 x}{\cos^2 x} dx = \int \tan^2 x dx \\ = \int (\sec^2 x - 1) dx = \tan x - x + C$$

$$20) I = \int \frac{(2 - 3\sin x)}{\cos^2 x} dx = \int \frac{2}{\cos^2 x} - \frac{3\sin x}{\cos^2 x} dx \\ = \int (2\sec^2 x - 3\sec x \tan x) dx = 2\tan x - 3\sec x + C$$

$$21) I = \int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right) dx = \frac{2}{3} x^{3/2} + 2\sqrt{x} + C$$

$$22) \text{ If } \frac{d}{dx}[f(x)] = 4x^3 - \frac{3}{x^4} \text{ such that } f(2) = 0, \text{ then } f(x) = ?$$

$$\text{Gn: } \frac{d}{dx}[f(x)] = 4x^3 - \frac{3}{x^4}$$

Integrating on both sides,

$$\int \frac{d}{dx}[f(x)] dx = \int \left(4x^3 - \frac{3}{x^4} \right) dx$$

$$f(x) = \frac{4x^4}{4} - \frac{3x^{-3}}{-3} + C = x^4 + x^{-3} + C$$

$$f(2) = \frac{16}{1} + \frac{1}{8} + C$$

$$0 = \frac{129}{8} + C$$

$$\Rightarrow C = -\frac{129}{8}$$

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$\therefore f(x) = x^4 + \frac{1}{x^3} - \frac{129}{8}$

Eg 4) Find the antiderivative F of f defined by $f(x) = 4x^3 - 6$ where $F(0) = 3$.

Given: $f(x) = 4x^3 - 6$

Integrating on both the sides,

$$\int f(x) \cdot dx = \int (4x^3 - 6) \cdot dx$$

$$F(x) = \frac{4x^4}{4} - 6x + C \rightarrow \textcircled{1}$$

$$F(0) = 0 - 0 + C$$

$$3 = C$$

$\therefore \textcircled{1} \Rightarrow F(x) = x^4 - 6x + 3$

EXERCISE 7.2

Eg 5) i) $I = \int \sin mx \cdot dx = -\frac{\cos mx}{m} + C$

ii) $I = \int 2x \cdot \sin(x^2+1) \cdot dx$

$= \int \sin t \cdot dt$

$= -\cos t + C$

$= -\cos(x^2+1) + C$

iii) ** $I = \int \frac{\tan^4 \sqrt{x} \cdot \sec^2 \sqrt{x}}{\sqrt{x}} \cdot dx$

$\tan \sqrt{x} = t$

$\frac{\sec^2 \sqrt{x}}{2\sqrt{x}} \cdot dx = dt$

$= \int 2t^3 \cdot dt$

$= 2 \frac{t^4}{4} + C$

$= \frac{2}{5} \tan^5 \sqrt{x} + C$

$$\begin{aligned} \text{iv) } I &= \int \frac{\sin(\tan^{-1}x)}{1+x^2} dx \\ \text{Let } \tan^{-1}x &= t \quad \frac{1}{1+x^2} dx = dt \\ I &= \int \sin t \cdot dt \\ &= -\cos t + C \\ &= -\cos(\tan^{-1}x) + C \end{aligned}$$

$$\begin{aligned} \text{Eg 6) i) } I &= \int \sin^3 x \cdot \cos^2 x \cdot dx \\ &= \int \sin^2 x \cdot \cos^2 x \cdot \sin x \cdot dx \\ &= \int (1 - \cos^2 x) \cos^2 x \sin x \cdot dx \\ &= \int (1 - t^2) t^2 (-1) dt \quad \text{Let } \cos x = t \quad -\sin x \cdot dx = -dt \\ &= -\int (t^2 - t^4) dt \\ &= -\left[\frac{t^3}{3} - \frac{t^5}{5} \right] + C \\ &= -\frac{\cos^3 x}{3} + \frac{\cos^5 x}{5} + C \end{aligned}$$

$$\begin{aligned} \text{ii) } I &= \int \frac{\sin x}{\sin(x+a)} dx \\ \text{Let } x+a &= t \quad dx = dt \\ I &= \int \frac{\sin(t-a)}{\sin t} dt \\ &= \int \frac{\sin t \cos a - \cos t \sin a}{\sin t} dt \\ &= \int (\cos a - \sin a \cdot \cot t) dt \\ &= \cos a \cdot t - \sin a \cdot \log(\sin t) + C \\ &= \cos a (x+a) - \sin a \cdot \log(\sin(x+a)) + C \end{aligned}$$

$$\begin{aligned} \text{iii) } I &= \int \frac{dx}{1+\tan x} = \int \frac{dx}{1+\frac{\sin x}{\cos x}} = \int \frac{\cos x \cdot dx}{\cos x + \sin x} \\ &= \frac{1}{2} \int \frac{2 \cos x \cdot dx}{\cos x + \sin x} = \frac{1}{2} \int \frac{(\cos x + \cos x)}{\cos x + \sin x} dx \end{aligned}$$

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$$\begin{aligned}
 I &= \frac{1}{2} \int \frac{\cos x + \sin x - \sin x + \cos x}{\cos x + \sin x} dx \\
 &= \frac{1}{2} \int \frac{(\cos x + \sin x) + (\cos x - \sin x)}{\cos x + \sin x} dx \\
 &= \frac{1}{2} \int 1 + \frac{\cos x - \sin x}{\cos x + \sin x} dx \\
 &= \frac{1}{2} \left[x + \log(\cos x + \sin x) \right] + C
 \end{aligned}$$

32) $I = \int \frac{dx}{1 + \cot x} = \int \frac{dx}{1 + \frac{\cos x}{\sin x}} = \int \frac{\sin x \cdot dx}{\sin x + \cos x} = \frac{2}{2}$

$$\begin{aligned}
 &= \frac{1}{2} \int \frac{2 \sin x \cdot dx}{\sin x + \cos x} = \frac{1}{2} \int \frac{\sin x + \sin x + \cos x - \cos x}{\sin x + \cos x} dx \\
 &= \frac{1}{2} \int \frac{(\sin x) + \cos x - (\cos x - \sin x)}{\sin x + \cos x} dx \\
 &= \frac{1}{2} \int 1 - \frac{(\cos x - \sin x)}{\sin x + \cos x} dx \\
 &= \frac{1}{2} \left[x - \log(\cos x + \sin x) \right] + C
 \end{aligned}$$

1) $I = \int \frac{2x}{1+x^2} dx = \log|1+x^2| + C$

2) $I = \int \frac{(\log x)^2}{x} dx$ $\log x = t$
 $\frac{1}{x} dx = dt$

$$\begin{aligned}
 &= \int t^2 \cdot dt \\
 &= \frac{t^3}{3} + C = \frac{(\log x)^3}{3} + C
 \end{aligned}$$

** 3) $I = \int \frac{1}{x + x \log x} dx = \int \frac{1}{x(1 + \log x)} dx$

$$\begin{aligned}
 &= \log|1 + \log x| + C
 \end{aligned}$$

4) $I = \int \sin x \cdot \sin(\cos x) \cdot dx$ $\sin(\cos x) = t$ $I =$
 $= -\int \sin t \cdot dt$ $\sin(\cos x) = t$ $-\sin x \cdot dx = dt$
 $= -(-\cos t) + c$ $\sin(\cos x) = t$ $\cos t = \cos(\cos x)$
 $= \cos t + c$ $\sin(\cos x) = t$ $\cos t = \cos(\cos x)$
 $= \cos(\cos x) + c$

5) $I = \int \sin(ax+b) \cdot \cos(ax+b) \cdot dx$ $\sin(ax+b) = t$ $I =$
 $= -\int \frac{1}{a} t \cdot dt$ $\sin(ax+b) = t$ $-\sin(ax+b) \cdot a \cdot dx = dt$
 $= -\frac{1}{a} \cdot \frac{t^2}{2} + c$ $\sin(ax+b) = t$ $\sin(ax+b) \cdot dx = -\frac{1}{a} dt$
 $= -\frac{1}{2a} \cdot \cos^2(ax+b) + c$ $\sin(ax+b) = t$ $\cos^2(ax+b) = 1 - \sin^2(ax+b)$
 $I = \int \sin(ax+b) \cdot \cos(ax+b) \cdot dx$ $\sin x \cdot \cos x = \frac{\sin 2x}{2}$
 $= \int \frac{\sin 2(ax+b)}{2} \cdot dx$ $\sin 2(ax+b) = \sin 2u$ $\frac{1}{2} \int \sin 2u \cdot du$
 $= \frac{1}{2} \left[-\frac{\cos(2ax+2b)}{2a} \right] + c$ $\sin 2u = \sin 2x$ $\frac{1}{2} \int \sin 2u \cdot du$
 $= -\frac{\cos(2ax+2b)}{4a} + c$ $\sin 2u = \sin 2x$ $\frac{1}{2} \int \sin 2u \cdot du$

6) $I = \int \sqrt{ax+b} \cdot dx$ $\sqrt{ax+b} = t$ $I =$
 $= \frac{2}{3} \frac{(ax+b)^{3/2}}{a} + c$ $\sqrt{ax+b} = t$ $\frac{1}{2} \int t^2 \cdot dt$
 $= \frac{2}{3} \frac{(ax+b)^{3/2}}{a} + c$ $\sqrt{ax+b} = t$ $\frac{1}{2} \int t^2 \cdot dt$

* 7) $I = \int x \cdot \sqrt{x+2} \cdot dx$ $x+2 = t$ $I =$
 $= \int (t-2) \sqrt{t} \cdot dt$ $x+2 = t$ $dx = dt$ $I =$
 $= \int t^{3/2} - 2t^{1/2} \cdot dt$ $x+2 = t$ $dx = dt$ $I =$
 $= \frac{2}{5} t^{5/2} - 2 \cdot \frac{2}{3} t^{3/2} + c$ $x+2 = t$ $dx = dt$ $I =$

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$$= \frac{2}{5} (x+2)^{5/2} - \frac{4}{3} (x+2)^{3/2} + C$$

8) $I = \int x \sqrt{1+2x^2} \cdot dx$

$$1+2x^2 = t$$

$$4x \cdot dx = dt$$

$$x \cdot dx = \frac{1}{4} \cdot dt$$

$$= \frac{1}{4} \int \sqrt{t} \cdot dt$$

$$= \frac{1}{4} \cdot \frac{2}{3} \cdot t^{3/2} + C$$

$$= \frac{1}{6} (1+2x^2)^{3/2} + C$$

9) $I = \int (4x+2) \sqrt{x^2+x+1} \cdot dx$

$$x^2+x+1 = t$$

$$(2x+1) \cdot dx = dt$$

$$(4x+2) \cdot dx = 2dt$$

$$= \int 2dt \sqrt{t}$$

$$= 2 \cdot \frac{2}{3} t^{3/2} + C$$

$$= \frac{4}{3} (x^2+x+1)^{3/2} + C$$

10) $I = \int \frac{1}{x-\sqrt{x}} \cdot dx = \int \frac{1}{\sqrt{x}(\sqrt{x}-1)} \cdot dx$

$$\sqrt{x}-1 = t$$

$$\frac{1}{2\sqrt{x}} \cdot dx = dt$$

$$\frac{1}{\sqrt{x}} \cdot dx = 2dt$$

$$= \int \frac{2dt}{t} = 2 \log t + C$$

$$= 2 \log (\sqrt{x}-1) + C$$

** 11) (a) $I = \int \frac{x}{\sqrt{x+4}} \cdot dx = \int \frac{(x+4)-4}{\sqrt{x+4}} \cdot dx$

$$= \int \left(\sqrt{x+4} - \frac{4}{\sqrt{x+4}} \right) \cdot dx$$

$$= \frac{2}{3} (x+4)^{3/2} - 4(2\sqrt{x+4}) + C$$

(b) $I = \int \frac{x}{\sqrt{x+5}} \cdot dx = \int \frac{(x+5)-5}{\sqrt{x+5}} \cdot dx$

$$= \int \left(\sqrt{x+5} - \frac{5}{\sqrt{x+5}} \right) \cdot dx = \frac{2}{3} (x+5)^{3/2} - 5(2\sqrt{x+5}) + C$$

$$(c) I = \int \frac{x}{x+s} dx = \int \frac{(x+s)-s}{(x+s)} dx = \int \left(1 - \frac{s}{x+s}\right) dx = x - s \log(x+s) + C$$

$$12] I = \int (x^3-1)^{1/3} \cdot x^5 \cdot dx$$

$$= \int (x^3-1)^{1/3} \cdot x^2 \cdot x^3 \cdot dx$$

$$= \int t^{1/3} \cdot (t+1) \cdot \frac{1}{3} \cdot dt$$

$$= \frac{1}{3} \left(\frac{3}{7} t^{7/3} + \frac{3}{4} t^{4/3} \right) + C$$

$$= \frac{1}{7} (x^3-1)^{7/3} + \frac{1}{4} (x^3-1)^{4/3} + C$$

$$13] I = \int \frac{x^2 \cdot dx}{(2+3x^3)^3}$$

$$= \frac{1}{9} \frac{dt}{t^3}$$

$$= \frac{1}{9} \frac{t^{-2}}{-2} + C$$

$$= -\frac{1}{18} \frac{1}{(2+3x^3)^2} + C$$

$$14] I = \int \frac{1}{x(\log x)^m} \cdot dx$$

$$= \int \frac{dt}{t^m}$$

$$= \frac{t^{-m+1}}{-m+1} + C$$

$$= \frac{1}{(-m+1)} (\log x)^{-m+1} + C$$

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15)
$$I = \int \frac{x \cdot dx}{9 - 4x^2}$$

$$9 - 4x^2 = t$$

$$-8x \cdot dx = dt$$

$$x \cdot dx = \frac{dt}{-8}$$

$$= -\frac{1}{8} \int \frac{dt}{t}$$

$$= -\frac{1}{8} \log t + c$$

$$= -\frac{1}{8} \log(9 - 4x^2) + c$$

16)
$$I = \int e^{2x+3} dx = \frac{e^{2x+3}}{2} + c$$

17)
$$I = \int \frac{x}{(e^x)^2} \cdot dx$$

$$x^2 = t$$

$$2x \cdot dx = dt$$

$$x \cdot dx = \frac{dt}{2}$$

$$= \frac{1}{2} \int \frac{dt}{e^t}$$

$$= \frac{1}{2} \int e^{-t} \cdot dt$$

$$= \frac{1}{2} \frac{e^{-t}}{-1} + c = -\frac{1}{2} e^{-x^2} + c$$

18)
$$I = \int \frac{e^{\tan^{-1}x}}{1+x^2} \cdot dx$$

$$\tan^{-1}x = t$$

$$\frac{1}{1+x^2} \cdot dx = dt$$

$$I = \int e^t \cdot dt = e^t + c$$

$$= e^{\tan^{-1}x} + c$$

19)
$$I = \int \frac{e^{2x} - 1}{e^{2x} + 1} \cdot dx$$

$$\div \text{num \& den by } e^x$$

$$= \int \frac{e^x - \frac{1}{e^x}}{e^x + \frac{1}{e^x}} \cdot dx$$

$$= \log \left| e^x + \frac{1}{e^x} \right| + c$$

$$\begin{aligned}
 20) \quad I &= \int \frac{e^{2x} - e^{-2x}}{e^{2x} + e^{-2x}} dx \\
 &= \frac{1}{2} \int \frac{2(e^{2x} - e^{-2x})}{e^{2x} + e^{-2x}} dx \\
 &= \frac{1}{2} \log |e^{2x} + e^{-2x}| + C
 \end{aligned}$$

$$\begin{aligned}
 21) \quad I &= \int \tan^2(2x-3) dx = \int \sec^2(2x-3) - 1 \cdot dx \\
 &= \frac{\tan(2x-3)}{2} - x + C
 \end{aligned}$$

$$22) \quad I = \int \sec^2(7-4x) \cdot dx = \frac{\tan(7-4x)}{-4} + C$$

$$\begin{aligned}
 23) \quad I &= \int \frac{\sin^{-1}x}{\sqrt{1-x^2}} dx \quad \sin^{-1}x = t \\
 &= \int t \cdot dt \quad \frac{1}{\sqrt{1-x^2}} dx = dt \\
 &= \frac{t^2}{2} + C \\
 &= \frac{(\sin^{-1}x)^2}{2} + C
 \end{aligned}$$

$$\begin{aligned}
 24) \quad I &= \int \frac{(2\cos x - 3\sin x) dx}{6\cos x + 4\sin x} = \frac{1}{2} \int \frac{2\cos x - 3\sin x}{3\cos x + 2\sin x} dx \\
 &= \frac{1}{2} \log |3\cos x + 2\sin x| + C
 \end{aligned}$$

$$\begin{aligned}
 25) \quad I &= \int \frac{1}{\cos^2 x (1 - \tan x)^2} dx = \int \frac{\sec^2 x \cdot dx}{(1 - \tan x)^2} \\
 &= - \int \frac{dt}{t^2} = \frac{1}{t} + C = \frac{1}{1 - \tan x} + C
 \end{aligned}$$

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26] a) $I = \int \frac{\cos \sqrt{x}}{\sqrt{x}} dx$ $\sqrt{x} = t$
 $\frac{1}{2\sqrt{x}} dx = dt$
 $\frac{1}{\sqrt{x}} dx = 2 dt$
 $= \int \cos t \cdot 2 dt$
 $= \sin t \cdot 2 + C$
 $= 2 \sin \sqrt{x} + C$

(b) $I = \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$ $\sqrt{x} = t$
 $\frac{1}{2\sqrt{x}} dx = dt$
 $\frac{1}{\sqrt{x}} dx = 2 dt$
 $= \int \sin t \cdot 2 dt$
 $= -\cos t \cdot 2 + C$
 $= -2 \cos \sqrt{x} + C$

27] $I = \int \sqrt{\sin 2x} \cdot \cos 2x \cdot dx$ $\sin 2x = t$
 $2 \cos 2x \cdot dx = dt$
 $\cos 2x \cdot dx = \frac{dt}{2}$
 $= \frac{1}{2} \int \sqrt{t} \cdot dt$
 $= \frac{1}{2} \cdot \frac{2}{3} t^{3/2} + C$
 $= \frac{1}{3} (\sin 2x)^{3/2} + C$

28] $I = \int \frac{\cos x}{\sqrt{1+\sin x}} \cdot dx$ $1 + \sin x = t$
 $\cos x \cdot dx = dt$
 $= \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C$
 $= 2\sqrt{1+\sin x} + C$

29] $I = \int \cot x \cdot \log(\sin x) \cdot dx$ $\log(\sin x) = t$
 $\frac{\cos x}{\sin x} \cdot dx = dt$
 $\cot x \cdot dx = dt$
 $= \int t \cdot dt$
 $= \frac{t^2}{2} + C$
 $= \frac{(\log(\sin x))^2}{2} + C$

$$30] \quad I = \int \frac{\sin x}{1+\cos x} dx = - \int \frac{(-\sin x) dx}{1+\cos x} = - \int \frac{d(1+\cos x)}{1+\cos x} = - \log|1+\cos x| + C = \log\left(\frac{1}{1+\cos x}\right) + C$$

$$31] \quad I = \int \frac{\sin x \cdot dx}{(1+\cos x)^2} \quad \begin{aligned} 1+\cos x &= t \\ -\sin x \cdot dx &= dt \end{aligned}$$

$$= \int \frac{-dt}{t^2} = \frac{1}{t} + C = \frac{1}{1+\cos x} + C$$

$$32] \quad I = \int \frac{1}{1-\tan x} dx = \int \frac{\cos x \cdot dx}{\cos x - \sin x} = \frac{1}{2} \int \frac{2 \cos x dx}{\cos x - \sin x}$$

$$= \frac{1}{2} \int \frac{(\cos x - \sin x) - (-\cos x - \sin x)}{\cos x - \sin x} dx$$

$$= \frac{1}{2} \int \left[1 - \frac{(-\cos x - \sin x)}{\cos x - \sin x} \right] dx$$

$$= \frac{1}{2} \left[x + \log|\cos x - \sin x| \right] + C$$

$$** 34] \quad I = \int \frac{\sqrt{\tan x} dx}{\sin x \cdot \cos x} \quad \text{Multiply \& denominator by } \cos x$$

$$= \int \frac{\sqrt{\tan x} \cdot dx}{\left(\frac{\sin x}{\cos x}\right) \cos x \cdot \cos x} = \int \frac{\sqrt{\tan x} \cdot \sec^2 x \cdot dx}{\tan x}$$

$$= \int \frac{\sec^2 x \cdot dx}{\sqrt{\tan x}} \quad \begin{aligned} \tan x &= t \\ \sec^2 x \cdot dx &= dt \end{aligned}$$

$$= \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C = 2\sqrt{\tan x} + C$$

35) $I = \int \frac{(1+\log x)^2}{x} dx$ $1+\log x = t$
 $\frac{1}{x} \cdot dx = dt$
 $= \int t^2 \cdot dt$
 $= \frac{t^3}{3} + C = \frac{(1+\log x)^3}{3} + C$

36) $I = \int \frac{(x+1)(x+\log x)^2}{x} dx$
 $= \int \frac{(x+1)}{x} \cdot (x+\log x)^2 \cdot dx = \int \left(1 + \frac{1}{x}\right) (x+\log x)^2 \cdot dx$
 $= \int t^2 \cdot dt$ $\left| \begin{array}{l} x+\log x = t \\ \left(1 + \frac{1}{x}\right) dx = dt \end{array} \right.$
 $= \frac{t^3}{3} + C = \frac{1}{3} (x+\log x)^3 + C$

37) $I = \int \frac{x^3 \sin(\tan^{-1} x)}{1+x^8} dx$ $\tan^{-1} x = t$
 $\frac{4x^3}{1+x^8} \cdot dx = dt$
 $= \frac{1}{4} \int \sin t \cdot dt$
 $= -\frac{1}{4} \cos t + C$
 $= -\frac{1}{4} \cos(\tan^{-1} x) + C$

38) $I = \int \frac{10x^9 + 10^x \log_e 10}{x^{10} + 10^x} dx = \log |x^{10} + 10^x| + C$

39) $I = \int \frac{dx}{\sin^2 x \cdot \cos^2 x} = \int \frac{\cos^2 x + \sin^2 x}{\sin^2 x \cdot \cos^2 x} \cdot dx$
 $= \int \sec^2 x + \csc^2 x \cdot dx$
 $= \tan x - \cot x + C$

EXERCISE 7.3

1) $I = \int \sin^2(2x+5) \cdot dx = \int \frac{1 - \cos(4x+10)}{2} \cdot dx$ $\sin^2 x = \frac{1 - \cos 2x}{2}$
 $= \frac{1}{2} \left[x - \frac{\sin(4x+10)}{4} \right] + C$

2) (a) $I = \int \sin 3x \cdot \cos 4x \cdot dx$ $SA \cdot CB = \frac{1}{2} [S(A+B) + S(A-B)]$
 $= \frac{1}{2} \int (\sin 7x - \sin x) \cdot dx$ $CA \cdot SB = \frac{1}{2} [S(A+B) - S(A-B)]$
 $= \frac{1}{2} \left[-\frac{\cos 7x}{7} + \cos x \right] + C$ $CA \cdot CB = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$
 $SA \cdot SB = \frac{1}{2} [\cos(A+B) - \cos(A-B)]$

(b) $I = \int \cos 3x \cdot \sin 5x \cdot dx = \frac{1}{2} \int \sin 8x + \sin 2x \cdot dx$
 $= \frac{1}{2} \left[-\frac{\cos 8x}{8} - \frac{\cos 2x}{2} \right] + C$

(c) $I = \int \cos 3x \cdot \cos 5x \cdot dx = \frac{1}{2} \int \cos 8x + \cos 2x \cdot dx$
 $= \frac{1}{2} \left[\frac{\sin 8x}{8} + \frac{\sin 2x}{2} \right] + C$

(d) $I = \int \sin 6x \cdot \sin x \cdot dx = -\frac{1}{2} \int \cos 7x - \cos 5x \cdot dx$
 $= -\frac{1}{2} \left[\frac{\sin 7x}{7} - \frac{\sin 5x}{5} \right] + C$

3) $I = \int \cos 2x \cdot \cos 4x \cdot \cos 6x \cdot dx$
 $= \frac{1}{2} \int (\cos 8x + \cos 4x) \cos 4x \cdot dx$
 $= \frac{1}{2} \int (\cos 8x \cdot \cos 4x + \cos^2 4x) \cdot dx$ $\cos^2 x = \frac{1 + \cos 2x}{2}$
 $= \frac{1}{2} \int \left[\frac{1}{2} (\cos 12x + \cos 4x) + \frac{(1 + \cos 8x)}{2} \right] \cdot dx$
 $= \frac{1}{4} \left[\frac{\sin 12x}{12} + \frac{\sin 4x}{4} + x + \frac{\sin 8x}{8} \right] + C$

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4) $I = \int \sin^3(2x+1) \cdot dx$ $\sin^3 x = \frac{3\sin x - \sin 3x}{4}$
 $= \frac{1}{4} \int 3 \sin(2x+1) - \sin(6x+3) \cdot dx$
 $= \frac{1}{4} \left[\frac{-3 \cos(2x+1)}{2} + \frac{\cos(6x+3)}{6} \right] + C$

5) $I = \int \sin^3 x \cdot \cos^3 x \cdot dx = \int (\sin x \cdot \cos x)^3 \cdot dx = \int \left(\frac{\sin 2x}{2} \right)^3 \cdot dx$
 $= \frac{1}{8} \int \frac{3 \sin 2x - \sin 6x}{4} \cdot dx = \frac{1}{32} \left[\frac{-3 \cos 2x}{2} + \frac{\cos 6x}{6} \right] + C$

6) $I = \int \sin x \cdot \sin 2x \cdot \sin 3x \cdot dx$
 $= \frac{1}{2} \int (\cos 4x - \cos 2x) \sin 2x \cdot dx$
 $= -\frac{1}{2} \int (\cos 4x \cdot \sin 2x - \cos 2x \cdot \sin 2x) \cdot dx$
 $= -\frac{1}{2} \int \left[\frac{1}{2} (\sin 6x - \sin 2x) - \frac{\sin 4x}{2} \right] dx$
 $= -\frac{1}{4} \left[\frac{-\cos 6x}{6} + \frac{\cos 2x}{2} - \frac{\cos 4x}{4} \right] + C$

7) $I = \int \sin 4x \cdot \cos 8x \cdot \sin 8x \cdot dx$
 $I = -\frac{1}{2} \int (\cos 12x - \cos 4x) \cdot dx$
 $= -\frac{1}{2} \left[\frac{\sin 12x}{12} - \frac{\sin 4x}{4} \right] + C$

8) $I = \int \frac{1 - \cos x}{1 + \cos x} \cdot dx = \int \frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \cdot dx = \int \tan^2 \frac{x}{2} \cdot dx$
 $= \int (\sec^2 \frac{x}{2} - 1) \cdot dx = \frac{2 \tan \frac{x}{2}}{2} - x + C$

9) $I = \int \frac{\cos x \cdot dx}{1 + \cos x} = \int \frac{\cos x}{1 + \cos x} \cdot \frac{1 - \cos x}{1 - \cos x} \cdot dx$

$$\begin{aligned}
 I &= \int \frac{\cos x - \cos^2 x}{\sin^2 x} dx = \int (\operatorname{cosec} x \cdot \cot x - \cot^2 x) dx \\
 &= \int \operatorname{cosec} x \cdot \cot x - (\operatorname{cosec}^2 x - 1) dx \\
 &= -\operatorname{cosec} x + \cot x + x + C
 \end{aligned}$$

$$\begin{aligned}
 12) \quad I &= \int \frac{\sin^2 x}{1 + \cos x} dx = \int \frac{\sin^2 x}{1 + \cos x} \cdot \frac{1 - \cos x}{1 - \cos x} dx \\
 &= \int \frac{\sin^2 x (1 - \cos x)}{\sin^2 x} dx = \int (1 - \cos x) dx = x - \sin x + C
 \end{aligned}$$

$$\begin{aligned}
 10) \quad I &= \int \sin^4 x \cdot dx = \int (\sin^2 x)^2 \cdot dx = \int \left(\frac{1 - \cos 2x}{2} \right)^2 dx \\
 &= \frac{1}{4} \int (1 + \cos^2 2x - 2 \cos 2x) dx \\
 &= \frac{1}{4} \int \left(1 + \frac{(\cos 4x + 1)}{2} - 2 \cos 2x \right) dx \\
 &= \frac{1}{4} \int \left(\frac{3}{2} + \frac{\cos 4x}{2} - 2 \cos 2x \right) dx \\
 &= \frac{1}{4} \left[\frac{3x}{2} + \frac{\sin 4x}{8} - \frac{2 \sin 2x}{2} \right] + C
 \end{aligned}$$

$$\begin{aligned}
 11) \quad I &= \int \cos^4 2x \cdot dx = \int (\cos^2 2x)^2 \cdot dx = \int \left(\frac{1 + \cos 4x}{2} \right)^2 dx \\
 &= \frac{1}{4} \int (1 + \cos^2 4x + 2 \cos 4x) dx = \frac{1}{4} \int \left(1 + \frac{(\cos 8x + 1)}{2} + 2 \cos 4x \right) dx \\
 &= \frac{1}{4} \int \left(\frac{3}{2} + \frac{\cos 8x}{2} + 2 \cos 4x \right) dx \\
 &= \frac{1}{4} \left[\frac{3x}{2} + \frac{\sin 8x}{16} + \frac{2 \sin 4x}{4} \right] + C
 \end{aligned}$$

13)
$$I = \int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx = \int \frac{2(\cos^2 x - 1) - (2\cos^2 \alpha - 1)}{\cos x - \cos \alpha} dx$$

$$= \int \frac{2(\cos^2 x - \cos^2 \alpha) - 1 + 1}{\cos x - \cos \alpha} dx$$

$$= 2 \int \frac{(\cos x + \cos \alpha)(\cos x - \cos \alpha)}{(\cos x - \cos \alpha)} dx$$

$$= 2 (\sin x + \sin \alpha) + C$$

14)
$$I = \int \frac{\cos x - \sin x}{1 + \sin 2x} dx = \int \frac{\cos x - \sin x}{\cos^2 x + \sin^2 x + 2\sin x \cos x} dx$$

$$= \int \frac{\cos x - \sin x}{(\cos x + \sin x)^2} dx$$

$\cos x + \sin x = t$
 $(-\sin x + \cos x) dx = dt$

$$= \int \frac{dt}{t^2} = -\frac{1}{t} + C$$

$$= -\frac{1}{\cos x + \sin x} + C$$

15)
$$I = \int \tan^3 2x \cdot \sec 2x \cdot dx = \int \tan^2 2x \cdot \tan 2x \cdot \sec 2x \cdot dx$$

$$= \int (\sec^2 2x - 1) \tan 2x \cdot \sec 2x \cdot dx$$

$$= \int \sec^2 2x \cdot \sec 2x \cdot \tan 2x \cdot dx - \int \sec 2x \cdot \tan 2x \cdot dx$$

$\sec 2x = t$
 $2 \sec 2x \cdot \tan 2x \cdot dx = dt$

$$I = \frac{1}{2} \int t^2 \cdot dt - \frac{\sec 2x}{2} + C$$

$$= \frac{1}{2} \frac{t^3}{3} - \frac{\sec 2x}{2} + C$$

$$= \frac{1}{2} \left(\frac{\sec^3 2x}{3} - \frac{\sec 2x}{2} \right) + C$$

16)
$$I = \int \tan^4 x \cdot dx = \int (\tan^2 x)(\tan^2 x) dx$$

$$= \int (\sec^2 x - 1) \tan^2 x \cdot dx = \int \sec^2 x \cdot \tan^2 x - \tan^2 x \cdot dx$$

$$= \int \sec^2 x \cdot \tan^2 x \cdot dx - \int \tan^2 x \cdot dx$$

$\tan x = t$
 $\sec^2 x \cdot dx = dt$

$$= \int t^2 \cdot dt - \int (\sec^2 x - 1) dx$$

$$= \frac{t^3}{3} - \tan x + x + C$$

$$= \frac{\tan^3 x}{3} - \tan x + x + C$$

17)
$$I = \int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx = \int \frac{\sin x}{\cos^2 x} + \frac{\cos x}{\sin^2 x} dx$$

$$= \int \sec x \cdot \tan x + \operatorname{cosec} x \cdot \cot x \cdot dx$$

$$= \sec x - \operatorname{cosec} x + C$$

18)
$$I = \int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} dx = \int \frac{(1 - 2\sin^2 x) + 2\sin^2 x}{\cos^2 x} dx$$

$$= \int \frac{1}{\cos^2 x} \cdot dx = \int \sec^2 x \cdot dx = \tan x + C$$

19)
$$I = \int \frac{1}{\sin x \cdot \cos^3 x} \cdot dx$$
 Multiply and divide denominator by $\cos x$

$$= \int \frac{dx}{\frac{\sin x}{\cos x} \cdot \cos x \cdot \cos^2 x} = \int \frac{\sec^2 x \cdot \sec^2 x \cdot dx}{\tan x}$$

$$= \int \frac{(1 + \tan^2 x) \cdot \sec^2 x \cdot dx}{\tan x}$$

$\tan x = t$
 $\sec^2 x \cdot dx = dt$

$$I = \int \left(\frac{1+t^2}{t} \right) dt = \int \left(\frac{1}{t} + t \right) dt = \log t + \frac{t^2}{2} + C$$

$$= \log(\tan x) + \frac{\tan^2 x}{2} + C$$

$$20) I = \int \frac{\cos 2x \cdot dx}{(\cos x + \sin x)^2} = \int \frac{\cos^2 x - \sin^2 x}{(\cos x + \sin x)^2} dx$$

$$= \int \frac{(\cos x - \sin x)(\cos x + \sin x)}{(\cos x + \sin x)^2} dx = \int \frac{\cos x - \sin x}{\cos x + \sin x} dx$$

$$= \log |\cos x + \sin x| + C$$

$$21) I = \int \sin^{-1}(\cos x) \cdot dx = \int \sin^{-1}\left(\sin\left(\frac{\pi}{2} - x\right)\right) dx$$

$$= \int \frac{\pi}{2} - x dx = \frac{\pi x}{2} - \frac{x^2}{2} + C$$

$$22) I = \int \frac{dx}{\cos(x-a) \cos(x-b)} \quad \text{Multiply and divide by } \sin(b-a).$$

$$= \frac{1}{\sin(b-a)} \int \frac{\sin(b-a) dx}{\cos(x-a) \cos(x-b)} = \frac{1}{\sin(b-a)} \int \frac{\sin(b-a+x-x) dx}{\cos(x-a) \cos(x-b)}$$

$$= \frac{1}{\sin(b-a)} \int \frac{\sin[(x-a) - (x-b)] dx}{\cos(x-a) \cos(x-b)}$$

$$= \frac{1}{\sin(b-a)} \int \frac{\sin(x-a) \cdot \cos(x-b) - \cos(x-a) \sin(x-b)}{\cos(x-a) \cos(x-b)} dx$$

$$= \frac{1}{\sin(b-a)} \int \tan(x-a) - \tan(x-b) dx$$

$$= \frac{1}{\sin(b-a)} \left[\log |\sec(x-a)| - \log |\sec(x-b)| \right] + C$$

$$= \frac{1}{\sin(b-a)} \log \left| \frac{\sec(x-a)}{\sec(x-b)} \right| + C$$

$$23) \int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cdot \cos^2 x} dx = \int \sec^2 x - \operatorname{cosec}^2 x \cdot dx$$

$$= \tan x + \cot x + C$$

$$24) \int \frac{e^x(1+x) dx}{\cos^2(xe^x)} \quad \left[\begin{array}{l} xe^x = t \\ (xe^x + e^x) dx = dt \\ e^x(x+1) dx = dt \end{array} \right]$$

$$= \int \frac{dt}{\cos^2 t} = \int \sec^2 t dt = \tan t + C = \tan(xe^x) + C$$

Special Integrals:

$$I \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C \quad \left[\begin{array}{l} x = a \tan \theta \\ dx = a \sec^2 \theta \end{array} \right]$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C \quad \left[\begin{array}{l} \text{partial fractions on} \\ (a-x)(a+x) \end{array} \right]$$

$$\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C \quad \left[\begin{array}{l} \text{partial fractions on} \\ (x-a)(x+a) \end{array} \right]$$

$$II \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C \quad \left[\begin{array}{l} x = a \sin \theta \\ dx = a \cos \theta \end{array} \right]$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \log |x + \sqrt{a^2 + x^2}| + C \quad \left[\begin{array}{l} x = a \tan \theta \\ dx = a \sec^2 \theta \end{array} \right]$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log |x + \sqrt{x^2 - a^2}| + C \quad \left[\begin{array}{l} x = a \sec \theta \\ dx = a \sec \theta \tan \theta \end{array} \right]$$

$$III \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C \quad \left[\text{by parts} \right]$$

$$\int \sqrt{a^2 + x^2} dx = \frac{x}{2} \sqrt{a^2 + x^2} + \frac{a^2}{2} \log |x + \sqrt{a^2 + x^2}| + C$$

$$\int \sqrt{x^2 - a^2} dx = \frac{x}{2} \sqrt{x^2 - a^2} - \frac{a^2}{2} \log |x + \sqrt{x^2 - a^2}| + C$$

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PROOFS:

1] $\int \frac{1}{a^2+x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

LHS = I = $\int \frac{dx}{a^2+x^2}$ $x = a \tan \theta$; $\theta = \tan^{-1} \frac{x}{a}$
 $dx = a \sec^2 \theta \cdot d\theta$

$= \int \frac{a \sec^2 \theta \cdot d\theta}{a^2 + a^2 \tan^2 \theta}$

$= \int \frac{a \sec^2 \theta \cdot d\theta}{a^2 (1 + \tan^2 \theta)} = \frac{1}{a} \int \frac{\sec^2 \theta \cdot d\theta}{\sec^2 \theta}$

$= \frac{1}{a} \int 1 \cdot d\theta$

$= \frac{1}{a} \theta + C = \frac{1}{a} \tan^{-1} \frac{x}{a} + C = \text{RHS}$

$\therefore \int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$

2] $\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$

LHS = $\frac{1}{a^2-x^2} = \frac{1}{(a+x)(a-x)}$ $\frac{A}{a+x} + \frac{B}{a-x}$

$1 = A(a-x) + B(a+x)$

$x = -a : 1 = A(2a) \Rightarrow A = \frac{1}{2a}$

$x = a : 1 = B(2a) \Rightarrow B = \frac{1}{2a}$

$I = \int \frac{dx}{a^2-x^2} = \int \frac{1}{2a(a+x)} dx + \int \frac{1}{2a(a-x)} dx$

$= \frac{1}{2a} \left(\log |a+x| + \log |a-x| \right) + C$

$= \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$

$$3) \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$\frac{1}{x^2 - a^2} = \frac{1}{(x-a)(x+a)} = \frac{A}{x-a} + \frac{B}{x+a}$$

$$1 = A(x+a) + B(x-a)$$

put $x = a : 1 = A(2a) \Rightarrow A = \frac{1}{2a}$

put $x = -a : 1 = B(-2a) \Rightarrow B = -\frac{1}{2a}$

$$\therefore I = \int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \int \frac{1}{x-a} dx - \frac{1}{2a} \int \frac{1}{x+a} dx$$

$$= \frac{1}{2a} \left[\log|x-a| - \log|x+a| \right] + C$$

$$= \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$4) \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$$

$$I = \int \frac{1}{\sqrt{a^2 - x^2}} dx \quad x = a \sin \theta ; \theta = \sin^{-1} \frac{x}{a}$$

$$dx = a \cos \theta d\theta$$

$$= \int \frac{a \cos \theta d\theta}{\sqrt{a^2 - a^2 \sin^2 \theta}}$$

$$= \int \frac{a \cos \theta d\theta}{a \sqrt{1 - \sin^2 \theta}} = \int \frac{\cos \theta d\theta}{\cos \theta} = \int 1 d\theta$$

$$= \theta + C$$

$$= \sin^{-1} \frac{x}{a} + C$$

$$5) \int \frac{1}{\sqrt{a^2 + x^2}} dx = \log |x + \sqrt{a^2 + x^2}| + C$$

$$I = \int \frac{dx}{\sqrt{a^2 + x^2}} \quad x = a \tan \theta$$

$$\tan \theta = \frac{x}{a}$$

$$dx = a \sec^2 \theta d\theta$$

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$$\begin{aligned}
 I &= \int \frac{a \sec^2 \theta \, d\theta}{\sqrt{a^2 + a^2 \tan^2 \theta}} & \sec \theta &= \sqrt{1 + \tan^2 \theta} \\
 &= \int \frac{a \sec^2 \theta \, d\theta}{a \sqrt{1 + \tan^2 \theta}} & &= \sqrt{1 + \frac{x^2}{a^2}} \\
 &= \int \frac{\sec^2 \theta \, d\theta}{\sec \theta} & \sec \theta &= \frac{\sqrt{a^2 + x^2}}{a} \\
 &= \log |\sec \theta + \tan \theta| + C \\
 &= \log \left| \frac{\sqrt{a^2 + x^2}}{a} + \frac{x}{a} \right| + C \\
 &= \log \left| \frac{x + \sqrt{a^2 + x^2}}{a} \right| + C \\
 &= \log |x + \sqrt{a^2 + x^2}| - \log a + C \\
 \therefore \int \frac{1}{\sqrt{a^2 + x^2}} \, dx &= \log |x + \sqrt{a^2 + x^2}| + C
 \end{aligned}$$

5) $\int \frac{1}{\sqrt{x^2 - a^2}} \, dx = \log |x + \sqrt{x^2 - a^2}| + C$

$$\begin{aligned}
 I &= \int \frac{1}{\sqrt{x^2 - a^2}} \, dx & x &= a \sec \theta \\
 &= \int \frac{a \sec \theta \cdot \tan \theta \, d\theta}{\sqrt{a^2 \sec^2 \theta - a^2}} & \sec \theta &= \frac{x}{a} \\
 &= \int \frac{a \sec \theta \tan \theta \, d\theta}{a \sqrt{\sec^2 \theta - 1}} & \tan \theta &= \sqrt{\sec^2 \theta - 1} \\
 &= \int \frac{\sec \theta \tan \theta \, d\theta}{\tan \theta} & &= \sqrt{\frac{x^2}{a^2} - 1} \\
 &= \int \sec \theta \, d\theta & \tan \theta &= \sqrt{x^2 - a^2} \\
 &= \log |\sec \theta + \tan \theta| + C & dx &= a \sec \theta \cdot \tan \theta \, d\theta
 \end{aligned}$$

$$I = \log \left| \frac{x + \sqrt{x^2 - a^2}}{a} \right| + C$$

$$= \log \left| \frac{x + \sqrt{x^2 - a^2}}{a} \right| + C$$

$$= \log |x + \sqrt{x^2 - a^2}| - \log a + C$$

$$= \log |x + \sqrt{x^2 - a^2}| + C$$

7] Integration by parts: I L A T E

ITF Log Algebra Trigonometry

$$\int I \cdot \frac{d}{dx} U \cdot dx = I \cdot U - \int U \cdot \frac{d}{dx} (I) \cdot dx$$

Eg 1] $\int x \cdot e^x dx = x \cdot e^x - \int e^x \cdot 1 \cdot dx = x e^x - e^x + C$

Eg 2] $\int x \cdot \cos x \cdot dx = x \sin x - \int \sin x \cdot 1 \cdot dx = x \sin x + \cos x + C$

Eg 3] $\int x \cdot \sec^2 x \cdot dx = x \cdot \tan x - \int \tan x \cdot 1 \cdot dx = x \tan x - \log |\sec x| + C$

Eg 4] $\int x^2 \cdot e^{2x} \cdot dx = \frac{x^2 \cdot e^{2x}}{2} - \int \frac{e^{2x}}{2} \cdot 2x \cdot dx$

$$= \frac{x^2 \cdot e^{2x}}{2} - \left[x \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} \cdot 1 \cdot dx \right] + C$$

$$= \frac{x^2 \cdot e^{2x}}{2} - x \cdot \frac{e^{2x}}{2} + \frac{1}{2} \cdot \frac{e^{2x}}{2} + C$$

7] $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + C$

$$I = \int \sqrt{a^2 - x^2} \cdot 1 \cdot dx$$

$$= \sqrt{a^2 - x^2} \cdot x - \int x \cdot \frac{(-2x)}{2\sqrt{a^2 - x^2}} \cdot dx$$

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$$I = x\sqrt{a^2-x^2} - \left[\int \frac{(a^2-x^2)-a^2}{\sqrt{a^2-x^2}} dx \right]$$

$$I = x\sqrt{a^2-x^2} - \int \sqrt{a^2-x^2} dx + \int \frac{a^2}{\sqrt{a^2-x^2}} dx$$

$$I + I = x\sqrt{a^2-x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$2I = x\sqrt{a^2-x^2} + a^2 \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$I = \frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2} \sin^{-1}\left(\frac{x}{a}\right) + C$$

8) $\int \sqrt{a^2+x^2} dx = \frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \log|x + \sqrt{a^2+x^2}| + C$

$$I = \int \sqrt{a^2+x^2} \cdot 1 dx = \sqrt{a^2+x^2} \cdot x - \int x \cdot \frac{(2x)}{2\sqrt{a^2+x^2}} dx$$

$$I = x\sqrt{a^2+x^2} - \left[\int \frac{(a^2+x^2)-a^2}{\sqrt{a^2+x^2}} dx \right]$$

$$I = x\sqrt{a^2+x^2} - \int \sqrt{a^2+x^2} dx + a^2 \int \frac{1}{\sqrt{a^2+x^2}} dx$$

$$2I = x\sqrt{a^2+x^2} + a^2 \log|x + \sqrt{a^2+x^2}| + C$$

$$I = \frac{x}{2} \sqrt{a^2+x^2} + \frac{a^2}{2} \log|x + \sqrt{a^2+x^2}| + C$$

9) $\int \sqrt{x^2-a^2} dx = \frac{x}{2} \sqrt{x^2-a^2} - \frac{a^2}{2} \log|x + \sqrt{x^2-a^2}| + C$

$$I = \int \sqrt{x^2-a^2} \cdot 1 dx = \sqrt{x^2-a^2} \cdot x - \int x \cdot \frac{(2x)}{2\sqrt{x^2-a^2}} dx$$

$$I = x\sqrt{x^2-a^2} - \left[\int \frac{(x^2-a^2)+a^2}{\sqrt{x^2-a^2}} dx \right]$$

$$I = x\sqrt{x^2-a^2} - \int \sqrt{x^2-a^2} \cdot dx - a^2 \int \frac{dx}{\sqrt{x^2-a^2}} \quad x = T$$

$$2I = x\sqrt{x^2-a^2} - a^2 \log|x+\sqrt{x^2-a^2}| + C$$

$$I = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2} \log|x+\sqrt{x^2-a^2}| + C$$

Completing the square:

$$\begin{aligned} & ax^2 + bx + c \\ &= a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) \\ &= a\left\{\left(x + \frac{1}{2}\frac{b}{a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2}\right\} = a\left\{\left(x + \frac{b}{2a}\right)^2 + \left(\right)^2\right\} \end{aligned}$$

$$\begin{aligned} \text{Eg: } & 1 - x^2 - 2x^2 \\ &= -2\left\{x^2 + \frac{1}{2}x - \frac{1}{2}\right\} = -2\left\{\left(x + \frac{1}{4}\right)^2 - \frac{1}{2} - \frac{1}{16}\right\} \\ &= -2\left\{\left(x + \frac{1}{4}\right)^2 - \frac{9}{16}\right\} = -2\left\{\left(x + \frac{1}{4}\right)^2 - \left(\frac{3}{4}\right)^2\right\} \\ &= 2\left\{\left(\frac{3}{4}\right)^2 - \left(x + \frac{1}{4}\right)^2\right\} \end{aligned}$$

$$\begin{aligned} \text{Eg] } & x^2 + 2x + 2 \quad \text{Eg] } 1 + 2x + 3x^2 \\ &= (x+1)^2 + 2 - 1 = 3\left\{x^2 + \frac{2}{3}x + \frac{1}{3}\right\} \\ &= (x+1)^2 + 1^2 \\ &= 3\left\{\left(x + \frac{1}{3}\right)^2 + \frac{1}{3} - \frac{1}{9}\right\} \\ &= 3\left\{\left(x + \frac{1}{3}\right)^2 + \frac{2}{9}\right\} \\ &= 3\left\{\left(x + \frac{1}{3}\right)^2 + \left(\frac{\sqrt{2}}{3}\right)^2\right\} \end{aligned}$$

$$\begin{aligned} \text{Eg] } & (x-5)(x-4) = x^2 - 9x + 20 \\ &= \left(x - \frac{9}{2}\right)^2 + 20 - \frac{81}{4} \\ &= \left(x - \frac{9}{2}\right)^2 - \frac{1}{4} \\ &= \left(x - \frac{9}{2}\right)^2 - \left(\frac{1}{2}\right)^2 \end{aligned}$$

EXERCISE 7.4

Eg 8 i] $x^2 - 6x + 13$
 $= (x-3)^2 + 13 - 9$
 $= (x-3)^2 + 2^2$

(a) $\int \frac{dx}{x^2 - 6x + 13}$

$\int \frac{dx}{(x-3)^2 + 2^2} = \frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C$

(b) $\int \frac{dx}{\sqrt{x^2 - 6x + 13}} = \int \frac{dx}{\sqrt{(x-3)^2 + 2^2}} = \log |(x-3) + \sqrt{(x-3)^2 + 2^2}| + C$

(c) $\int \sqrt{x^2 - 6x + 13} dx = \int \sqrt{(x-3)^2 + 2^2} dx$
 $= \frac{x-3}{2} \sqrt{(x-3)^2 + 2} + \frac{4}{2} \log |(x-3) + \sqrt{(x-3)^2 + 2^2}| + C$
 $= \frac{x-3}{2} \sqrt{x^2 - 6x + 13} + 2 \log |(x-3) + \sqrt{x^2 - 6x + 13}| + C$

Eg 9 ii] $3x^2 - 13x - 10$

$= 3 \left\{ x^2 + \frac{13}{3}x - \frac{10}{3} \right\} = 3 \left\{ \left(x + \frac{13}{6} \right)^2 - \frac{10}{3} - \frac{169}{36} \right\}$

$= 3 \left\{ \left(x + \frac{13}{6} \right)^2 - \frac{289}{36} \right\} = 3 \left\{ \left(x + \frac{13}{6} \right)^2 - \left(\frac{17}{6} \right)^2 \right\}$

(a) $\int \frac{dx}{3x^2 - 13x - 10} = \frac{1}{3} \int \frac{dx}{\left(x + \frac{13}{6} \right)^2 - \left(\frac{17}{6} \right)^2}$

$= \frac{1}{3} \cdot \frac{1}{2 \left(\frac{17}{6} \right)} \log \left| \frac{x + \frac{13}{6} - \frac{17}{6}}{x + \frac{13}{6} + \frac{17}{6}} \right| + C$

$= \frac{1}{17} \log \left| \frac{6x - 4}{6x + 30} \right| + C$

$= \frac{1}{17} \log \left| \frac{3x - 2}{3x + 15} \right| + C$

$$1b) \int \frac{dx}{\sqrt{3x^2+13x-10}} = \frac{1}{\sqrt{3}} \int \frac{dx}{\left(x+\frac{13}{6}\right)^2 - \left(\frac{17}{6}\right)^2}$$

$$= \frac{1}{\sqrt{3}} \log \left| \left(x+\frac{13}{6}\right) + \sqrt{\left(x+\frac{13}{6}\right)^2 - \left(\frac{17}{6}\right)^2} \right| + C$$

$$\text{Eg 8] i) } \int \frac{dx}{x^2-16} = \int \frac{dx}{x^2-4^2} = \frac{1}{2(4)} \log \left| \frac{x-4}{x+4} \right| + C$$

$$\text{ii) } \int \frac{dx}{\sqrt{2x-x^2}} = \int \frac{dx}{\sqrt{1-(x-1)^2}} = \sin^{-1}(x-1) + C$$

$$\text{Eg 9] i) } \int \frac{dx}{x^2-6x+13} = \int \frac{dx}{(x-3)^2+2^2} = \frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C$$

$$\text{ii) } \int \frac{dx}{3x^2+13x-10}$$

$$= \frac{1}{3} \int \frac{dx}{\left(x+\frac{13}{6}\right)^2 - \left(\frac{17}{6}\right)^2}$$

$$= \frac{1}{3} \cdot \frac{6}{2(17)} \log \left| \frac{x+\frac{13}{6} - \frac{17}{6}}{x+\frac{13}{6} + \frac{17}{6}} \right| + C$$

$$= \frac{1}{17} \log \left| \frac{6x-4}{6x+30} \right| + C = \frac{1}{17} \log \left| \frac{3x-2}{3x+15} \right| + C$$

$$\text{iii) } \int \frac{dx}{\sqrt{5x^2-2x}} = \frac{1}{\sqrt{5}} \int \frac{dx}{\sqrt{\left(x-\frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^2}}$$

$$= \frac{1}{\sqrt{5}} \log \left| \left(x-\frac{1}{5}\right) + \sqrt{\left(x-\frac{1}{5}\right)^2 - \left(\frac{1}{5}\right)^2} \right| + C$$

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1) Ex 10.1 Q. 1) $I = \int \frac{3x^2}{x^6+1} dx = \int \frac{3x^2}{(x^3)^2+1^2} dx$ $\begin{matrix} \text{Let } x^3 = t \\ 3x^2 \cdot dx = dt \end{matrix}$

$$= \int \frac{dt}{t^2+1} = \frac{1}{1} \tan^{-1}\left(\frac{t}{1}\right) + C$$

$$= \tan^{-1}(x^3) + C$$

2) $I = \int \frac{1}{\sqrt{1+4x^2}} dx = \int \frac{dx}{\sqrt{1+(2x)^2}}$ $\begin{matrix} 2x = t \\ 2 \cdot dx = dt \\ dx = \frac{1}{2} dt \end{matrix}$

$$= \frac{1}{2} \int \frac{dt}{\sqrt{1+t^2}}$$

$$= \frac{1}{2} \log |t + \sqrt{1+t^2}| + C$$

$$= \frac{1}{2} \log |2x + \sqrt{1+4x^2}| + C$$

(or)

(a) $I = \int \frac{1}{\sqrt{1+4x^2}} dx = \int \frac{dx}{\sqrt{1+(2x)^2}} = \frac{\log |2x + \sqrt{1+4x^2}|}{2} + C$

(b) $I = \int \frac{dx}{\sqrt{1-9x^2}} = \int \frac{dx}{\sqrt{1-(3x)^2}} = \frac{\sin^{-1}(3x)}{3} + C$

(c) $I = \int \frac{dx}{\sqrt{9x^2-1}} = \int \frac{dx}{\sqrt{(3x)^2-1}} = \frac{\log |3x + \sqrt{9x^2-1}|}{3} + C$

3) $I = \int \frac{1}{\sqrt{(2-x)^2+1}} dx = \frac{\log |2-x + \sqrt{(2-x)^2+1}|}{1} + C$

4) $I = \int \frac{1}{\sqrt{9-25x^2}} dx = \int \frac{1}{\sqrt{3^2-(5x)^2}} dx = \frac{1}{5} \sin^{-1}\left(\frac{5x}{3}\right) + C$

5) $I = \int \frac{3x}{1+2x^4} dx = \int \frac{3x \cdot dx}{1^2+(\sqrt{2}x)^2}$ $\begin{matrix} \sqrt{2}x^2 = t \\ \sqrt{2} \cdot 2x \cdot dx = dt \\ x \cdot dx = \frac{1}{2\sqrt{2}} dt \end{matrix}$

$$= \int \frac{3 \cdot \frac{1}{2\sqrt{2}} dt}{1+t^2}$$

$$I = \frac{3}{2\sqrt{2}} \cdot \tan^{-1}(t) + C = \frac{3}{2\sqrt{2}} \cdot \tan^{-1}(\sqrt{2}x^2) + C$$

6)
$$I = \int \frac{x^2}{1-x^6} dx = \int \frac{x^2}{1-(x^3)^2} dx \quad \begin{matrix} x^3 = t \\ 3x^2 \cdot dx = dt \end{matrix}$$

$$= \int \frac{1}{3} \cdot \frac{dt}{1-t^2} = \frac{1}{3} \cdot \frac{1}{2(1)} \log \left| \frac{1+t}{1-t} \right| + C = \frac{1}{6} \log \left| \frac{1+x^3}{1-x^3} \right| + C$$

7)
$$I = \int \frac{(x-1)dx}{\sqrt{x^2-1}} = \int \frac{x \cdot dx}{\sqrt{x^2-1}} - \int \frac{1 \cdot dx}{\sqrt{x^2-1}}$$

$$x^2-1 = t \quad 2x \cdot dx = dt \quad x \cdot dx = \frac{1}{2} dt$$

$$I = \frac{1}{2} \int \frac{dt}{\sqrt{t}} - \log |x + \sqrt{x^2-1}| + C$$

$$= \frac{1}{2} \cdot 2\sqrt{x^2-1} - \log |x + \sqrt{x^2-1}| + C$$

8)
$$I = \int \frac{x^2}{\sqrt{x^6+a^6}} dx = \int \frac{x^2 \cdot dx}{\sqrt{(x^3)^2+(a^3)^2}} \quad \begin{matrix} x^3 = t \\ 3x^2 \cdot dx = dt \end{matrix}$$

$$= \int \frac{1}{3} \cdot \frac{dt}{\sqrt{t^2+(a^3)^2}} = \frac{1}{3} \int \frac{dt}{\sqrt{t^2+(a^3)^2}} \quad \begin{matrix} x^2 \cdot dx = \frac{1}{3} dt \end{matrix}$$

$$= \frac{1}{3} \cdot \log |t + \sqrt{t^2+(a^3)^2}| + C$$

$$= \frac{1}{3} \cdot \log |x^3 + \sqrt{x^6+a^6}| + C$$

9)
$$I = \int \frac{\sec^2 x \cdot dx}{\sqrt{\tan^2 x + 4}}$$

$$\tan x = t \quad \sec^2 x \cdot dx = dt$$

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$$\begin{aligned} I &= \int \frac{dt}{\sqrt{t^2+4}} = \log |t + \sqrt{t^2+4}| + C \\ &= \log |\tan x + \sqrt{\tan^2 x + 4}| + C \end{aligned}$$

$$\begin{aligned} 10) \quad I &= \int \frac{1 \cdot dx}{\sqrt{x^2+2x+2}} \\ &= \int \frac{dx}{\sqrt{(x+1)^2+1^2}} \\ &= \log |(x+1) + \sqrt{x^2+2x+2}| + C \end{aligned}$$

$$\begin{aligned} 11) \quad I &= \int \frac{1 \cdot dx}{9x^2+6x+5} = \frac{1}{9} \int \frac{dx}{(x+\frac{1}{3})^2+(\frac{2}{3})^2} \\ &= \frac{1}{9} \cdot \frac{3}{2} \tan^{-1} \left[\frac{x+\frac{1}{3}}{\frac{2}{3}} \right] + C \\ &= \frac{1}{6} \tan^{-1} \left(\frac{3x+1}{2} \right) + C \end{aligned}$$

$$\begin{aligned} 12) \quad I &= \int \frac{dx}{\sqrt{7-6x-x^2}} = \int \frac{dx}{\sqrt{4^2-(x+3)^2}} \\ &= \sin^{-1} \left(\frac{x+3}{4} \right) + C \end{aligned}$$

$$\begin{aligned} 13) \quad I &= \int \frac{1 \cdot dx}{\sqrt{(x-1)(x-2)}} = \int \frac{dx}{\sqrt{(x-\frac{3}{2})^2-(\frac{1}{2})^2}} \\ &= \log \left| (x-\frac{3}{2}) + \sqrt{(x-1)(x-2)} \right| + C \end{aligned}$$

$$14) \quad I = \int \frac{dx}{\sqrt{9+3x-x^2}}$$

$$\begin{aligned}
 I &= \int \frac{dx}{\sqrt{\left(\frac{\sqrt{41}}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2}} \\
 &= \sin^{-1} \left(\frac{x - \frac{3}{2}}{\frac{\sqrt{41}}{2}} \right) + C \\
 &= \sin^{-1} \left(\frac{2x - 3}{\sqrt{41}} \right) + C
 \end{aligned}$$

$$\begin{aligned}
 &-(x^2 - 3x - 8) \\
 &= \left[\left(x - \frac{3}{2}\right)^2 - 8 - \frac{9}{4} \right] \\
 &= \left[\left(x - \frac{3}{2}\right)^2 - \left(\frac{\sqrt{41}}{2}\right)^2 \right] \\
 &= \left(\frac{\sqrt{41}}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2
 \end{aligned}$$

$$\begin{aligned}
 15) \quad I &= \int \frac{dx}{\sqrt{(x-a)(x-b)}} = \int \frac{dx}{\sqrt{x^2 - (a+b)x + ab}} \\
 &= \int \frac{dx}{\sqrt{\left(x - \frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2}} \\
 &= \log \left| \left(x - \frac{a+b}{2}\right) + \sqrt{\left(x - \frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2} \right| + C
 \end{aligned}$$

$$\begin{aligned}
 24) \quad I &= \int \frac{dx}{x^2 + 2x + 2} = \int \frac{dx}{(x+1)^2 + 1^2} \\
 &= \tan^{-1}(x+1) + C
 \end{aligned}$$

$$\begin{aligned}
 25) \quad I &= \int \frac{dx}{\sqrt{9x - 4x^2}} = \int \frac{1}{2} \cdot \frac{dx}{\sqrt{\left(\frac{9}{8}\right)^2 - \left(x - \frac{9}{8}\right)^2}} \\
 &= \frac{1}{2} \sin^{-1} \left(\frac{8x-9}{9} \right) + C
 \end{aligned}$$



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EXERCISE 7.7.

Eg 23] $\int \sqrt{x^2 + 2x + 5} \cdot dx$

$$I = \int \sqrt{(x+1)^2 + 2^2} \cdot dx$$

$$= \frac{(x+1)}{2} \sqrt{(x+1)^2 + 2^2} + \frac{2^2}{2} \log |(x+1) + \sqrt{(x+1)^2 + 2^2}| + C$$

$$= \frac{(x+1)}{2} \sqrt{x^2 + 2x + 5} + 2 \log |(x+1) + \sqrt{x^2 + 2x + 5}| + C$$

* Eg 24] $\int \sqrt{3 - 2x - x^2} \cdot dx$

$$I = \int \sqrt{2^2 - (x+1)^2} \cdot dx$$

$$= \frac{(x+1)}{2} \sqrt{2^2 - (x+1)^2} + \frac{4^2}{2} \sin^{-1} \left(\frac{x+1}{2} \right) + C$$

$$= \frac{(x+1)}{2} \sqrt{3 - 2x - x^2} + 2 \sin^{-1} \left(\frac{x+1}{2} \right) + C$$

1] $\int \sqrt{4 - x^2} \cdot dx$

$$I = \int \sqrt{2^2 - x^2} = \frac{x}{2} \sqrt{4 - x^2} + \frac{4^2}{2} \sin^{-1} \left(\frac{x}{2} \right) + C$$

2] $\int \sqrt{4 - 4x^2} \cdot dx$

$$= \int \sqrt{1 - (2x)^2} \cdot dx$$

$$I = \frac{1}{2} \left[\frac{2x}{2} \sqrt{1 - 4x^2} + \frac{1}{2} \sin^{-1} (2x) \right] + C$$

$$I = \frac{1}{2} \left[x \sqrt{1 - 4x^2} + \frac{1}{2} \sin^{-1} (2x) \right] + C$$

$$I = 2 \int \left(\frac{1}{2} \right)^2 - x^2 \cdot dx = 2 \left[\frac{x}{2} \sqrt{1 - 4x^2} + \frac{1}{8} \sin^{-1} \frac{x}{\frac{1}{2}} \right] + C$$

$$I = \frac{x}{2} \sqrt{1 - 4x^2} + \frac{1}{4} \sin^{-1} (2x) + C$$

$$I = \frac{x}{2} \sqrt{1 - 4x^2} + \frac{1}{4} \sin^{-1} (2x) + C$$

$$3] \int \sqrt{x^2+4x+6} \, dx = \int (x+2)^2 + (\sqrt{2})^2 \cdot dx$$

$$= \frac{(x+2)}{2} \sqrt{x^2+4x+6} + \frac{x}{2} \log |(x+2) + \sqrt{x^2+4x+6}| + C$$

$$4] \int \sqrt{x^2+4x+1} \, dx = \int \sqrt{(x+2)^2 - (\sqrt{3})^2} \cdot dx$$

$$= \frac{(x+2)}{2} \sqrt{x^2+4x+1} - \frac{3}{2} \log |(x+2) + \sqrt{x^2+4x+1}| + C$$

$$5] \int \sqrt{1-4x-x^2} \, dx = \int \sqrt{(\sqrt{5})^2 - (x+2)^2} \, dx$$

$$= \frac{(x+2)}{2} \sqrt{1-4x-x^2} + \frac{5}{2} \sin^{-1} \left(\frac{x+2}{\sqrt{5}} \right) + C$$

$$6] \int \sqrt{x^2+4x-5} \, dx = \int \sqrt{(x+2)^2 - 3^2} \, dx$$

$$= \frac{(x+2)}{2} \sqrt{x^2+4x-5} - \frac{9}{2} \log |(x+2) + \sqrt{x^2+4x-5}| + C$$

$$7] \int \sqrt{1+3x-x^2} \, dx$$

$$= \int \sqrt{\left(\frac{\sqrt{13}}{2}\right)^2 - \left(x - \frac{3}{2}\right)^2} \, dx$$

$$= \frac{\left(x - \frac{3}{2}\right)}{2} \sqrt{1+3x-x^2} + \frac{13}{8} \sin^{-1} \left(\frac{x - \frac{3}{2}}{\frac{\sqrt{13}}{2}} \right) + C$$

$$= \frac{(2x-3)}{4} \sqrt{1+3x-x^2} + \frac{13}{8} \sin^{-1} \left(\frac{2x-3}{\sqrt{13}} \right) + C$$

$$8] \int \sqrt{x^2+3x} \, dx = \int \sqrt{\left(x + \frac{3}{2}\right)^2 - \left(\frac{3}{2}\right)^2} \, dx$$

$$= \frac{\left(x + \frac{3}{2}\right)}{2} \sqrt{x^2+3x} - \frac{9}{8} \log \left| \left(x + \frac{3}{2}\right) + \sqrt{x^2+3x} \right| + C$$

$$= \frac{(2x+3)}{4} \sqrt{x^2+3x} - \frac{9}{8} \log \left| \frac{2x+3}{2} + \sqrt{x^2+3x} \right| + C$$

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9] $I = \int \sqrt{1 + \frac{x^2}{9}} \cdot dx = \frac{1}{3} \int \sqrt{3^2 + x^2} \cdot dx$

$$= \frac{1}{3} \left[\frac{x}{2} \sqrt{9 + x^2} + \frac{9}{2} \log |x + \sqrt{9 + x^2}| \right] + C$$

10] $I = \int \sqrt{1 + x^2} \cdot dx$

$$= \frac{x}{2} \sqrt{1 + x^2} + \frac{1}{2} \log |x + \sqrt{1 + x^2}| + C$$

11] $I = \int \sqrt{x^2 - 8x + 7} \cdot dx$

$$= \int \sqrt{(x-4)^2 - (3)^2} \cdot dx$$

$$= \frac{(x-4)}{2} \sqrt{x^2 - 8x + 7} - \frac{9}{2} \log |x-4 + \sqrt{x^2 - 8x + 7}| + C$$

Integrals of the form : $\int \frac{(px+q) \cdot dx}{\sqrt{ax^2+bx+c}} / \int \frac{(px+q) \cdot dx}{ax^2+bx+c}$

→ Numerator = $A \frac{d}{dx}(de) + B$

→ Compare coefficients of x and constants to get values of A, B

→ Replace numerator, split and integrate.

→ $I = A \int \frac{\frac{d}{dx}(de)}{de} \cdot dx + B \int \frac{1}{\sqrt{ax^2+bx+c}} \cdot dx$

$$= A \log |de| + B (\text{complete the eq.})$$

Eg 10] i] $I = \int \frac{(x+2) dx}{2x^2+6x+5}$

$$N(x) = A \frac{d}{dx}(de) + B$$

$$(x+2) = A(4x+6) + B$$

coefficients of $x : 1 = 4A \Rightarrow A = \frac{1}{4}$

constants $2 = 6A + B$
 $2 - \frac{6}{4} = B \Rightarrow B = \frac{1}{2}$

$$I = \int \frac{A(4x+6) + B}{2x^2+6x+5} dx = \int \frac{\frac{1}{4}(4x+6) + \frac{1}{2}}{2x^2+6x+5} dx$$

$$= \frac{1}{4} \int \frac{(4x+6) dx}{2x^2+6x+5} + \frac{1}{2} \int \frac{dx}{2x^2+6x+5}$$

$2x^2+6x+5 = t$
 $(4x+6) \cdot dx = dt$

$$I = \frac{1}{4} \int \frac{dt}{t} + \frac{1}{2} \int \frac{dx}{(2)(x+\frac{3}{2})^2 + (\frac{1}{2})^2}$$

$$= \frac{1}{4} \log |2x^2+6x+5| + \frac{1}{4} \cdot \frac{2}{1} \tan^{-1} \left(\frac{x+\frac{3}{2}}{\frac{1}{2}} \right) + C$$

$$= \frac{1}{4} \log |2x^2+6x+5| + \frac{1}{2} \tan^{-1}(2x+3) + C$$

ii] $I = \int \frac{x+3}{\sqrt{5-4x+x^2}} dx$

Num = $A \frac{d(de)}{dx} + B$

$$x+3 = A(2x-4) + B$$

coeff of $x : 1 = 2A \Rightarrow A = \frac{1}{2}$

constants : $3 = -4(\frac{1}{2}) + B \Rightarrow B = 5$

$$I = \int \frac{\frac{1}{2}(2x-4) + 5}{\sqrt{5-4x+x^2}} dx = \frac{1}{2} \int \frac{(2x-4)}{\sqrt{5-4x+x^2}} dx + 5 \int \frac{dx}{\sqrt{5-4x+x^2}}$$

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$$5 - 4x + x^2 = t$$

$$(2x - 4) dx = dt$$

$$I = \frac{1}{2} \int \frac{dt}{\sqrt{t}} + 5 \int \frac{dx}{\sqrt{(x-2)^2 + 1^2}}$$

$$= \frac{1}{2} \cdot 2\sqrt{t} + 5 \int \log |(x-2) + \sqrt{x^2 - 4x + 5}| + C$$

$$= \sqrt{x^2 - 4x + 5} + 5 \log |(x-2) + \sqrt{x^2 - 4x + 5}| + C$$

EXERCISE 7.4

Special 16] $I = \int \frac{(4x+1) dx}{\sqrt{2x^2+x-3}}$

$$2x^2+x-3 = t$$

$$(4x+1) dx = dt$$

$$I = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C = 2\sqrt{2x^2+x-3} + C$$

17] $I = \int \frac{(x+2) dx}{\sqrt{x^2-1}}$

Numerator = $A \frac{d}{dx}(dx) + B$

$$x+2 = A(2x) + B$$

coeff of x : $1 = 2A \Rightarrow A = \frac{1}{2}$

constants : $2 = B$

$$I = \int \frac{\frac{1}{2}(2x) + 2}{\sqrt{x^2-1}} dx = \frac{1}{2} \int \frac{2x}{\sqrt{x^2-1}} dx + 2 \int \frac{dx}{\sqrt{x^2-1}}$$

$$\begin{aligned}
 & x^2 - 1 = t \\
 & (2x) dx = dt \\
 & I = \frac{1}{2} \int \frac{dt}{\sqrt{t}} + 2 \int \frac{dx}{\sqrt{x^2 - 1}} \\
 & = \frac{1}{2} (2\sqrt{t}) + 2 \log \left| \left(x - \frac{1}{x}\right) + \sqrt{x^2 - 1} \right| + C \\
 & = \sqrt{x^2 - 1} + 2 \log \left| \left(x - \frac{1}{x}\right) + \sqrt{x^2 - 1} \right| + C
 \end{aligned}$$

$$\begin{aligned}
 18) \int \frac{5x-2}{1+2x+3x^2} dx & \\
 3x^2+2x+1=t & \quad \text{Num} = A \frac{d}{dx}(t) + B \\
 (6x+2) dx = dt & \\
 5x-2 = A(6x+2) + B & \\
 \text{coefficients of } x: 5 = 6A \Rightarrow A = \frac{5}{6} & \\
 \text{constants: } -2 = 2A + B & \\
 -2 - 2\left(\frac{5}{6}\right) = B & \\
 B = -\frac{11}{3} & \\
 I = \int \frac{\frac{5}{6}(6x+2) - \frac{11}{3}}{1+2x+3x^2} dx & = \frac{5}{6} \int \frac{6x+2}{1+2x+3x^2} dx - \frac{11}{3} \int \frac{dx}{1+2x+3x^2} \\
 1+2x+3x^2=t & \quad 3x^2+2x+1 \\
 (6x+2) dx = dt & \quad 3\left(x^2+\frac{2}{3}x+\frac{1}{3}\right) \\
 & \quad 3\left[\left(x+\frac{1}{3}\right)^2+\frac{1}{3}-\frac{1}{9}\right] \\
 & \quad 3\left[\left(x+\frac{1}{3}\right)^2+\frac{2}{9}\right] \\
 & \quad 3\left[\left(x+\frac{1}{3}\right)^2+\left(\frac{\sqrt{2}}{3}\right)^2\right] \\
 & \quad = 3\left[\left(x+\frac{1}{3}\right)^2+\left(\frac{\sqrt{2}}{3}\right)^2\right] \\
 I = \frac{5}{6} \int \frac{dt}{t} - \frac{11}{3} \int \frac{dx}{\left(x+\frac{1}{3}\right)^2+\left(\frac{\sqrt{2}}{3}\right)^2} & \\
 = \frac{5}{6} \log|t| - \frac{11}{3} \cdot \frac{1}{\frac{\sqrt{2}}{3}} \tan^{-1} \left(\frac{x+\frac{1}{3}}{\frac{\sqrt{2}}{3}} \right) + C & \\
 = \frac{5}{6} \log|3x^2+2x+1| - \frac{11}{3\sqrt{2}} \tan^{-1} \left(\frac{3x+1}{\sqrt{2}} \right) + C &
 \end{aligned}$$

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19] $\int \frac{6x+7}{\sqrt{(x-5)(x-9)}} dx = \int \frac{(6x+7) dx}{\sqrt{x^2-9x+20}}$

Num = $A \frac{d}{dx}(de) + B$

$6x+7 = A(2x-9) + B$

coeff of x : $6 = 2A \Rightarrow A = 3$

const: $7 = -27 + B \Rightarrow B = 34$

$I = \int \frac{3(2x-9) + 34}{\sqrt{x^2-9x+20}} dx = 3 \int \frac{(2x-9) dx}{\sqrt{x^2-9x+20}} + 34 \int \frac{dx}{\sqrt{x^2-9x+20}}$

$x^2-9x+20 = t$
 $(2x-9) dx = dt$

$(x-\frac{9}{2})^2 + 20 - \frac{81}{4}$
 $(x-\frac{9}{2})^2 - (\frac{1}{2})^2$ $A = s+x$

$I = 3 \int \frac{dt}{\sqrt{t}} + 34 \int \frac{dx}{\sqrt{(x-\frac{9}{2})^2 - (\frac{1}{2})^2}}$

$I = 3(2\sqrt{t}) + 34 \log \left| (x-\frac{9}{2}) + \sqrt{(x-\frac{9}{2})^2 - (\frac{1}{2})^2} \right| + C$

$= 6\sqrt{x^2-9x+20} + 34 \log \left| \frac{(2x-9) + \sqrt{x^2-9x+20}}{2} \right| + C$

22] $\int \frac{(x+3)}{x^2-2x-5} dx$

Num = $A \frac{d}{dx}(de) + B$

$x+3 = A(2x-2) + B$

coeff of x : $1 = 2A \Rightarrow A = \frac{1}{2}$

constants: $3 = -2A + B \Rightarrow 3+1 = B = 4$

$I = \int \frac{\frac{1}{2}(2x-2) + 4}{x^2-2x-5} dx = \frac{1}{2} \int \frac{(2x-2) dx}{x^2-2x-5} + 4 \int \frac{dx}{x^2-2x-5}$

$x^2-2x-5 = t$
 $(2x-2) dx = dt$

$x^2-2x-5 = (x-1)^2 - 6$
 $(x-1)^2 - (\sqrt{6})^2$ $A = s+x$

$\frac{1}{t} = A \Rightarrow AS = -1$
 $1 = 8 - 5 \Rightarrow 3 + AS = 9$

$$\begin{aligned}
 I &= \frac{1}{2} \int \frac{dt}{t} + 4 \int \frac{dx}{(x-1)^2 - (\sqrt{6})^2} \\
 &= \frac{1}{2} \log\left(\frac{t}{2}\right) + 4 \cdot \frac{1}{2\sqrt{6}} \log \left| \frac{(x-1)-\sqrt{6}}{(x-1)+\sqrt{6}} \right| + C \\
 &= \frac{1}{2} \log|x^2-2x-5| + \frac{2}{\sqrt{6}} \log \left| \frac{x-1-\sqrt{6}}{x-1+\sqrt{6}} \right| + C
 \end{aligned}$$

$$\begin{aligned}
 20) \int \frac{(x+2) dx}{\sqrt{4x-x^2}} \\
 \text{Num} &= A \frac{d}{dx}(4x-x^2) + B(x-2) \\
 x+2 &= A(4-2x) + B(x-2) \\
 \text{coeff of } x: 1 &= -2A \Rightarrow A = -\frac{1}{2} \\
 \text{constant: } 2 &= 4A+B \Rightarrow 2+2=B=4 \\
 I &= \int \frac{-\frac{1}{2}(4-2x)+4}{\sqrt{4x-x^2}} dx = -\frac{1}{2} \int \frac{(4-2x)}{\sqrt{4x-x^2}} dx + 4 \int \frac{dx}{\sqrt{4x-x^2}}
 \end{aligned}$$

$$\begin{aligned}
 4x-x^2 &= t \\
 (4-2x)dx &= dt \\
 I &= -\frac{1}{2} \int \frac{dt}{\sqrt{t}} + 4 \int \frac{dx}{\sqrt{2^2-(x-2)^2}} \\
 &= -\frac{1}{2} (2\sqrt{t}) + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C \\
 &= -\sqrt{4x-x^2} + 4 \sin^{-1}\left(\frac{x-2}{2}\right) + C
 \end{aligned}$$

$$\begin{aligned}
 21) \int \frac{(x+2) dx}{\sqrt{x^2+2x+3}} \\
 \text{Num} &= A \frac{d}{dx}(x^2+2x+3) + B(1-x) \\
 x+2 &= A(2x+2) + B(1-x) \\
 \text{coeff of } x: 1 &= 2A \Rightarrow A = \frac{1}{2} \\
 \text{const}^n: 2 &= 2A+B \Rightarrow 2-1=B=1
 \end{aligned}$$

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$$I = \int \frac{\frac{1}{2}(2x+2) + 1}{\sqrt{x^2+2x+3}} dx = \frac{1}{2} \int \frac{(2x+2) dx}{\sqrt{x^2+2x+3}} + \int \frac{dx}{\sqrt{x^2+2x+3}}$$

$$\begin{aligned} x^2+2x+3 &= t \\ (2x+2)dx &= dt \end{aligned} \quad \left| \begin{aligned} x^2+2x+3 &= (x+1)^2+2 \\ &= (x+1)^2+(\sqrt{2})^2 \end{aligned} \right.$$

$$I = \frac{1}{2} \int \frac{dt}{\sqrt{t}} + \int \frac{dx}{\sqrt{(x+1)^2+(\sqrt{2})^2}}$$

$$= \frac{1}{2}(2\sqrt{t}) + \log |(x+1) + \sqrt{(x+1)^2+(\sqrt{2})^2}| + C$$

$$= \sqrt{x^2+2x+3} + \log |(x+1) + \sqrt{x^2+2x+3}| + C$$

23) $\int \frac{5x+3}{\sqrt{x^2+4x+10}} dx$

$$5x+3 = A(2x+4) + B$$

coeff of x : $5 = 2A \Rightarrow A = \frac{5}{2}$

constⁿ: $3 = 4A + B \Rightarrow 3 - 10 = B \Rightarrow B = -7$

$$I = \int \frac{\frac{5}{2}(2x+4) - 7}{\sqrt{x^2+4x+10}} dx = \frac{5}{2} \int \frac{(2x+4) dx}{\sqrt{x^2+4x+10}} - 7 \int \frac{dx}{\sqrt{x^2+4x+10}}$$

$$\begin{aligned} x^2+4x+10 &= t \\ (2x+4)dx &= dt \end{aligned} \quad \left| \begin{aligned} x^2+4x+10 &= (x+2)^2+6 \\ &= (x+2)^2+(\sqrt{6})^2 \end{aligned} \right.$$

$$I = \frac{5}{2} \int \frac{dt}{\sqrt{t}} - 7 \int \frac{dx}{\sqrt{(x+2)^2+(\sqrt{6})^2}}$$

$$= \frac{5}{2}(2\sqrt{t}) - 7 \log |(x+2) + \sqrt{(x+2)^2+(\sqrt{6})^2}| + C$$

$$= 5\sqrt{x^2+4x+10} - 7 \log |(x+2) + \sqrt{x^2+4x+10}| + C$$

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5) $I = \int x \cdot \log_2 x \cdot dx = \log_2 x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{2x} \cdot dx$
 $= \frac{x^2}{2} \log_2 x - \frac{1}{2} \cdot \frac{x^2}{2} + C$

6) $I = \int x^2 \cdot \log x \cdot dx = \log x \cdot \frac{x^3}{3} - \int \frac{x^3}{3} \cdot \frac{1}{x} \cdot dx$
 $= \frac{x^3}{3} \log x - \frac{1}{3} \cdot \frac{x^3}{3} + C$

7) $I = \int x \cdot \sin^{-1} x \cdot dx = \sin^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{\sqrt{1-x^2}} \cdot dx$
 $= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \int \frac{(1-x^2)-1}{\sqrt{1-x^2}} \cdot dx$
 $= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \int \sqrt{1-x^2} \cdot dx - \frac{1}{2} \int \frac{dx}{\sqrt{1-x^2}}$
 $= \frac{x^2}{2} \sin^{-1} x + \frac{1}{2} \left[\frac{x \sqrt{1-x^2}}{2} + \frac{1}{2} \sin^{-1}(x) \right] - \frac{1}{2} \sin^{-1} x + C$
 $= \frac{x^2}{2} \sin^{-1} x + \frac{x \sqrt{1-x^2}}{4} - \frac{1}{4} \sin^{-1} x + C$

8) $I = \int x \cdot \tan^{-1} x \cdot dx = \tan^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{(1+x^2)} \cdot dx$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{(x^2+1)-1}{1+x^2} \cdot dx$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{dx}{1+x^2} + \frac{1}{2} \int \frac{1}{1+x^2} \cdot dx$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} x + \frac{1}{2} \tan^{-1} x + C$

9) $I = \int x \cdot \cos^{-1} x \cdot dx = \cos^{-1} x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \left(\frac{-1}{\sqrt{1-x^2}} \right) \cdot dx$
 $= \frac{x^2}{2} \cos^{-1} x + \frac{1}{2} \int \frac{(1-x^2)-1}{\sqrt{1-x^2}} \cdot dx$

$$\begin{aligned}
 &= \frac{x^2}{2} \cos^{-1} x - \frac{1}{2} \int \sqrt{1-x^2} dx + \frac{1}{2} \int \frac{1}{\sqrt{1-x^2}} dx \quad [2] \\
 &= \frac{x^2}{2} \cos^{-1} x - \frac{1}{2} \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \log \sin^{-1} x \right] + \frac{1}{2} \sin^{-1} x + C \\
 &= \frac{x^2}{2} \cos^{-1} x - \frac{x}{4} \sqrt{1-x^2} + \frac{1}{4} \sin^{-1} x + C \quad [3]
 \end{aligned}$$

13] $I = \int \tan^{-1} x \cdot dx = \tan^{-1} x \cdot x - \int \frac{x}{1+x^2} dx$

$$\begin{aligned}
 &= \tan^{-1} x \cdot x - \frac{1}{2} \int \frac{2x}{1+x^2} dx \quad (\text{adjusting}) \\
 &= x \tan^{-1} x + \frac{1}{2} \log |1+x^2| + C
 \end{aligned}$$

14] $I = \int x (\log x)^2 dx = (\log x)^2 \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{2 \log x}{x} dx$

$$\begin{aligned}
 &= (\log x)^2 \cdot \frac{x^2}{2} - \int x \cdot \log x \cdot dx \\
 &= (\log x)^2 \cdot \frac{x^2}{2} - \left[\log x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \right] \\
 &= (\log x)^2 \cdot \frac{x^2}{2} - \frac{x^2}{2} \log x + \frac{1}{2} \int x \cdot dx \\
 &= \frac{x^2}{2} (\log x)^2 - \frac{x^2}{2} \log x + \frac{x^2}{4} + C
 \end{aligned}$$

15] $I = \int (x^2+1) \log x \cdot dx = \log x \left(\frac{x^3}{3} + x \right) - \int \left(\frac{x^3}{3} + x \right) \frac{1}{x} dx$

$$\begin{aligned}
 &= \left(\frac{x^3}{3} + x \right) \log x - \int \left(\frac{x^2}{3} + 1 \right) dx \\
 &= \left(\frac{x^3}{3} + x \right) \log x - \left(\frac{x^3}{9} + x \right) + C
 \end{aligned}$$

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NOTE : (a) $\int e^{ax} \cdot \sin bx \cdot dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx] + c$

(b) $\int e^{ax} \cdot \cos bx \cdot dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx] + c$

21) $I = \int e^{2x} \cdot \sin x \cdot dx = \frac{e^{2x}}{4+1} [2 \sin x - \cos x] + c$

$I = \int e^{2x} \cdot \sin x \cdot dx = -e^{2x} \cos x + \int \cos x \cdot 2e^{2x} \cdot dx$
 $= -e^{2x} \cos x + 2 \left[e^{2x} \cdot \sin x - \int \sin x \cdot 2e^{2x} \cdot dx \right]$
 $= -e^{2x} \cos x + 2 \left[e^{2x} \sin x - 2 \int \sin x \cdot e^{2x} \cdot dx \right]$
 $= -e^{2x} \cos x + 2e^{2x} \sin x - 4 \int \sin x \cdot e^{2x} \cdot dx + c$

$5I = -e^{2x} \cos x + 2e^{2x} \sin x + c$
 $I = \frac{e^{2x}}{5} (2 \sin x - \cos x) + c$

$I = \int e^x \cdot \sin x \cdot dx = -e^x \cos x + \int \cos x \cdot e^x \cdot dx$
 $= -e^x \cos x + \left[e^x \cdot \sin x - \int \sin x \cdot e^x \cdot dx \right] + c$
 $= -e^x \cos x + e^x \sin x - I$
 $2I = -e^x \cos x + e^x \sin x + c$
 $I = \frac{e^x}{2} [-\cos x + \sin x] + c$

~~$I = \int e^{2x} \cdot \cos 3x \cdot dx = \frac{e^{2x}}{3} \sin 3x - \int \sin 3x \cdot 2e^{2x} \cdot dx$
 $= \frac{e^{2x}}{3} \sin 3x - 2 \left[e^{2x} \left(-\frac{\cos 3x}{3} \right) + \int \frac{\cos 3x}{3} \cdot 2e^{2x} \cdot dx \right]$
 $= \frac{e^{2x}}{3} \sin 3x + \frac{2}{3} e^{2x} \cos 3x - \frac{4}{3} I + c$
 $\frac{7}{3} I = \frac{e^{2x}}{3} \left[\sin 3x + 2 \cos 3x \right] + c$~~

$$\begin{aligned}
 I &= \int e^{3x} \cdot \sin 4x \cdot dx = \frac{e^{3x}}{25} [3 \sin 4x - 4 \cos 4x] + C \\
 &= -e^{3x} \frac{\cos 4x}{4} + \int \frac{\cos 4x}{4} \cdot 3e^{3x} \cdot dx \\
 &= -e^{3x} \frac{\cos 4x}{4} + \frac{3}{4} \left[e^{3x} \cdot \sin 4x - \int \sin 4x \cdot 3e^{3x} dx \right] \\
 &= -e^{3x} \frac{\cos 4x}{4} + \frac{3}{16} e^{3x} \sin 4x - \frac{9}{16} \int e^{3x} \cdot \sin 4x \cdot dx \\
 I + \frac{9}{16} I &= e^{3x} \left[\frac{3}{16} \sin 4x - \frac{1}{4} \cos 4x \right] + C \\
 \frac{25I}{16} &= \frac{e^{3x}}{16} [3 \sin 4x - 4 \cos 4x] + C \\
 I &= \frac{e^{3x}}{25} [3 \sin 4x - 4 \cos 4x] + C
 \end{aligned}$$

Property Under By parts :

$$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$$

Proof : LHS = $\int e^x [f(x) + f'(x)] dx$

$$\begin{aligned}
 I &= \int e^x f(x) dx + \int e^x f'(x) dx \\
 &= \int e^x f(x) dx + e^x f(x) - \int f(x) \cdot e^x dx + C \\
 &= e^x f(x) + C = \text{RHS}
 \end{aligned}$$

16) $I = \int e^x (\sin x + \cos x) dx = e^x \sin x + C$

24) $I = \int e^x \sec x (1 + \tan x) dx = \int e^x [\sec x + \sec x \cdot \tan x] dx$

$$= e^x \sec x + C$$

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Eg 22] (a) $I = \int e^x \left[\tan^{-1} x + \frac{1}{1+x^2} \right] dx = e^x \tan^{-1} x + c$ (or)

(b) $I = \int \frac{(x^2+1)e^x dx}{(x+1)^2} = \int e^x \left[\frac{(x+1)^2 - 2x}{(x+1)^2} \right] dx$

$$= \int e^x \left[1 - 2 \frac{x}{(x+1)^2} \right] dx$$

$$= \int e^x dx - 2 \int e^x \cdot \frac{x}{(x+1)^2} dx$$

$$= e^x - 2 \left[\int e^x \cdot \frac{(x+1) - 1}{(x+1)^2} dx \right]$$

$$= e^x - 2 \left[\int e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right) dx \right]$$

$$= e^x - 2 \frac{e^x}{x+1} + c$$

17 *] $I = \int \frac{x \cdot e^x}{(1+x)^2} dx$

$$= \int e^x \left(\frac{(x+1) - 1}{(1+x)^2} \right) dx = \int e^x \left(\frac{1}{x+1} - \frac{1}{(x+1)^2} \right) dx$$

$$= \frac{e^x}{x+1} + c$$

18] $I = \int e^x \frac{(1+\sin x) dx}{1+\cos x} = e^x \frac{(1+2\sin \frac{x}{2})}{1+\cos x}$

$$= \int e^x \left[\frac{1+2\sin \frac{x}{2} \cdot \cos \frac{x}{2}}{2\cos^2 \frac{x}{2}} \right] dx = \int e^x \left[\frac{1}{2} \sec^2 \frac{x}{2} + \tan \frac{x}{2} \right] dx$$

$$= e^x \tan \frac{x}{2} + c$$

19] $I = \int e^x \frac{(x-1) dx}{x^2} = \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) dx$

$$= \frac{e^x}{x} + c$$

$$\begin{aligned}
 20) \quad I &= \int \frac{(x-3)e^x}{(x-1)^3} dx = \int \frac{(x-1-2)e^x}{(x-1)^3} dx = I_1 + I_2 \\
 &= \int e^x \left(\frac{1}{(x-1)^2} - \frac{2}{(x-1)^3} \right) dx \\
 &= \frac{e^x}{(x-1)^2} + C
 \end{aligned}$$

$$\begin{aligned}
 22) \quad I &= \int \sin^{-1} \left(\frac{2x}{1+x^2} \right) dx \quad x = \tan \theta \quad \theta = \tan^{-1} x \\
 &\quad dx = \sec^2 \theta \cdot d\theta \\
 &= \int \sin^{-1}(\sin 2\theta) \sec^2 \theta \cdot d\theta \\
 &= \int 2\theta \cdot \sec^2 \theta \cdot d\theta = 2 \left[\theta \tan \theta - \int \tan \theta \cdot d\theta \right] \\
 &= 2\theta \tan \theta - 2 \log(\sec \theta) + C \\
 &= 2x \tan^{-1} x - 2 \log(\sec(\tan^{-1} x)) + C
 \end{aligned}$$

Integration by Partial Fractions - Rules:

- * If the denominator has linear factors, take one constant for every factor.

$$\frac{2x-3}{x(x^2-1)} = \frac{2x-3}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

- * For every repeated linear factor of the form $(ax+b)^n$ take 'n' no. of constants each time increasing the power.

$$\frac{2x-3}{x(x+1)^2} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2}$$

- * If single repeated linear factor in denominator then put it as y.
 $\therefore 3x+1 = y \Rightarrow x = \frac{y-1}{3}$

$$\frac{2(y-1)-9}{3y^3} = \frac{2y-11}{3y^3}$$

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* If the denominator has quadratic polynomial as ax^2+bx+c or ax^2+c , use 2 constants simultaneously as $Ax+B$.

Eg: $\frac{2x-3}{x(x+1)^2(x^2+1)} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{(x+1)^2} + \frac{(Dx+E)}{x^2+1}$
LF RLF GF

NOTE: If in $\frac{f(x)}{g(x)}$, degree of numerator $\geq$ degree of denominator, then it is improper fraction (IF).

Divide and reduce IF as $\text{IF} = \text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}}$

Eg: $\frac{9}{2}$ $nv > de \Rightarrow \text{IF}$ $1 = A \cdot \frac{2}{2} = A \cdot 1 \Rightarrow A = 1$

$\frac{9}{2} = 4 + \frac{1}{2}$

EXERCISE 7.5

1) $\int \frac{x}{(x+1)(x+2)} dx$

$\frac{x}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$

$x = -1 : A = -1$
 $x = -2 : B = 2$

$I = \int \left(\frac{-1}{x+1} + \frac{2}{x+2} \right) dx = -\log(x+1) + 2\log(x+2) + C$
 $= \log \left| \frac{(x+2)^2}{(x+1)} \right| + C$

2) $\int \frac{1}{x^2-9} dx$

$\frac{1}{x^2-9} = \frac{1}{(x+3)(x-3)} = \frac{A}{x-3} + \frac{B}{x+3}$

$\frac{1}{x^2-9} = \frac{1}{(x+3)(x-3)} = \frac{A}{x-3} + \frac{B}{x+3}$

$I = \int \frac{1}{6(x-3)} dx + \int \frac{-1}{6(x+3)} dx$

$$I = \frac{1}{6} [\log(x-3) - \log(x+3)] + C$$

$$= \frac{1}{6} \log \left| \frac{x-3}{x+3} \right| + C$$

$$3) \int \frac{3x-1}{(x-1)(x-2)(x-3)} dx$$

$$\frac{3x-1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

$$x=1: 2 = 3x-1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

$$x=1: 2 = 2A \Rightarrow A=1$$

$$x=2: 5 = -B \Rightarrow B=-5$$

$$x=3: 8 = 2C \Rightarrow C=4$$

$$I = \int \frac{dx}{x-1} + \frac{-5}{5} \int \frac{dx}{x-2} + 4 \int \frac{dx}{x-3}$$

$$= \log|x-1| - 5 \log|x-2| + 4 \log|x-3| + C$$

$$4) \int \frac{x}{(x-1)(x-2)(x-3)} dx$$

$$\frac{x}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$$

$$x = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$$

$$\text{put } x=1: 1 = 2A \Rightarrow A = \frac{1}{2}$$

$$\text{put } x=2: 2 = -B \Rightarrow B = -2$$

$$\text{put } x=3: 3 = 2C \Rightarrow C = \frac{3}{2}$$

$$I = \frac{1}{2} \int \frac{dx}{x-1} - 2 \int \frac{dx}{x-2} + \frac{3}{2} \int \frac{dx}{x-3} = I$$

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$$I = \frac{1}{2} \log(x-1) - 2 \log(x-2) + \frac{3}{2} \log(x-3) + C$$

5] $I = \int \frac{2x \cdot dx}{x^2+3x+2} = \int \frac{2x \cdot dx}{(x+1)(x+2)}$

$$\frac{2x}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$2x = A(x+2) + B(x+1)$$

put $x = -1$: $-2 = A \Rightarrow A = -2$

put $x = -2$: $-4 = B \Rightarrow B = 4$

$$I = \int \frac{-2 \cdot dx}{x+1} + \int \frac{4 \cdot dx}{x+2}$$

$$= -2 \int \frac{dx}{x+1} + 4 \int \frac{dx}{x+2}$$

$$= -2 \log|x+1| + 4 \log|x+2| + C$$

$$= 2 \left[2 \log|x+2| - \log|x+1| \right] + C = 2 \log \left| \frac{(x+2)^2}{(x+1)} \right| + C$$

6] $I = \int \frac{1-x^2}{x(1-2x)} dx$

$$\frac{1-x^2}{x(1-2x)} = \frac{1}{x} + \frac{1}{2} \left(\frac{2x}{x(1-2x)} \right) \rightarrow \text{①}$$

$$\frac{2-x}{x(1-2x)} = \frac{A}{x} + \frac{B}{1-2x}$$

$$2-x = A(1-2x) + Bx$$

$$2-x = A - 2Ax + Bx$$

$$2-x = A + (B-2A)x$$

$$\begin{cases} A = 2 \\ B-2A = -1 \end{cases} \Rightarrow \begin{cases} A = 2 \\ B = 3 \end{cases}$$

$$\frac{2-x}{x(1-2x)} = \frac{2}{x} + \frac{3}{1-2x}$$

$$I = \int \frac{2}{x} dx + \int \frac{3}{1-2x} dx$$

$$= 2 \log|x| - \frac{3}{2} \log|1-2x| + C$$

$$\begin{aligned}
 x=0: A &= 2 \left(\frac{2}{1} \right) \\
 x=\frac{1}{2}: B &= 3 \left(\frac{3 \cdot 2}{1} \right) \\
 \textcircled{1} \Rightarrow \frac{1-x^2}{x(1-2x)} &= \frac{1}{2} + \frac{1}{2} \left(\frac{2}{x} + \frac{3}{1-2x} \right) \\
 \therefore I &= \int \frac{1-x^2}{x(1-2x)} dx = \frac{1}{2} \int \left(1 + \frac{2}{x} + \frac{3}{1-2x} \right) dx \\
 &= \frac{1}{2} \left[x + 2 \log x + \frac{3 \log (1-2x)}{2} \right] + C
 \end{aligned}$$

$$\begin{aligned}
 8) I &= \int \frac{x}{(x-1)^2(x+2)} dx \\
 \frac{x}{(x+2)(x-1)^2} &= \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2} \\
 x &= A(x-1)^2 + B(x-1)(x+2) + C(x+2) \\
 x=-2: -2 &= A(9) \Rightarrow A = -\frac{2}{9} \\
 x=1: 1 &= C(3) \Rightarrow C = \frac{1}{3} \\
 \text{coeff of } x^2: 0 &= A+B \Rightarrow B = \frac{2}{9} \\
 I &= \int \frac{-\frac{2}{9}}{x+2} dx + \frac{2}{9} \int \frac{1}{x-1} dx + \frac{1}{3} \int \frac{1}{(x-1)^2} dx \\
 &= -\frac{2}{9} \log |x+2| + \frac{2}{9} \log |x-1| + \frac{1}{3} \left(-\frac{1}{x-1} \right) + C \\
 &= \frac{2}{9} \log \left| \frac{x-1}{x+2} \right| - \frac{1}{3(x-1)} + C
 \end{aligned}$$

$$\begin{aligned}
 1) I &= \int \frac{x}{(x^2+1)(x-1)} dx \\
 \frac{x}{(x^2+1)(x-1)} &= \frac{A}{x-1} + \frac{Bx+C}{x^2+1} \\
 x &= A(x^2+1) + (Bx+C)(x-1) \\
 x=1: 1 &= A(2) \Rightarrow A = \frac{1}{2} \\
 \text{coeff of } x^2: 0 &= A+B \Rightarrow B = -\frac{1}{2}
 \end{aligned}$$

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coeff of x : $1 = C - B \Rightarrow C = \frac{1}{2}$

$$I = \frac{1}{2} \int \frac{dx}{x-1} + \frac{1}{2} \int \frac{-x+1}{x^2+1} dx$$

$$= \frac{1}{2} \log|x-1| + \frac{1}{2} \int \frac{1}{x^2+1} dx - \frac{1}{2} \int \frac{x}{x^2+1} dx$$

$$= \frac{1}{2} \log|x-1| + \frac{1}{2} \tan^{-1} x - \frac{1}{2} \int \frac{dt}{t}$$

$$= \frac{1}{2} \log|x-1| + \frac{1}{2} \tan^{-1} x - \frac{1}{2} \log t + C$$

$$= \frac{1}{2} \left(\log|x-1| + \tan^{-1} x - \log \sqrt{x^2+1} \right) + C$$

9) $\int \frac{(3x+5)dx}{x^3-x^2-x+1} = \int \frac{(3x+5)dx}{x^2(x-1)(x+1)}$

$$\frac{3x+5}{(x^2-1)(x-1)} = \frac{3x+5}{(x-1)(x+1)(x-1)} = \frac{3x+5}{(x+1)(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

$$3x+5 = A(x-1) + B + C(x+1)$$

$x=1$: $8 = 2C \Rightarrow C = 4$

$x=-1$: $2 = 4A \Rightarrow A = \frac{1}{2}$

coeff of x^2 : $0 = A+B \Rightarrow B = -\frac{1}{2}$

$$I = \frac{1}{2} \int \frac{dx}{x-1} - \frac{1}{2} \int \frac{dx}{(x-1)^2} + 4 \int \frac{dx}{x+1}$$

$$= \frac{1}{2} \left(\log|x-1| - \log|x-1| \right) + \frac{4}{x-1} + C$$

$$= \frac{1}{2} \log \left| \frac{x+1}{x-1} \right| + \frac{4}{x-1} + C$$

$$10) \int \frac{2x-3}{(x^2-1)(2x+3)} dx$$

$$\frac{2x-3}{(x^2-1)(2x+3)} = \frac{2x-3}{(x+1)(x-1)(2x+3)} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{2x+3}$$

$$2x-3 = A(x-1)(2x+3) + B(x+1)(2x+3) + C(x+1)(x-1)$$

put $x = -1$ $-5 = -2A \Rightarrow A = \frac{5}{2}$

put $x = 1$ $-1 = 10B \Rightarrow B = -\frac{1}{10}$

put $x = -\frac{3}{2}$ $-6 = \frac{5}{4}C \Rightarrow C = -\frac{24}{5}$

$$I = \frac{5}{2} \int \frac{dx}{x+1} - \frac{1}{10} \int \frac{dx}{x-1} - \frac{24}{5} \int \frac{dx}{2x+3}$$

$$= \frac{5}{2} \log|x+1| - \frac{1}{10} \log|x-1| - \frac{12}{5} \log|2x+3|$$

$$= \frac{5}{2} \log|x+1| - \frac{1}{10} \log|x-1| - \frac{12}{5} \log|2x+3|$$

$$11) I = \int \frac{5x}{(x+1)(x^2-4)} dx = \int \frac{5x}{(x+1)(x+2)(x-2)} dx = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x-2}$$

$$5x = A(x+2)(x-2) + B(x+1)(x-2) + C(x+1)(x+2)$$

put $x = -1$ $-5 = -3A \Rightarrow A = \frac{5}{3}$

$x = -2$ $-10 = 4B \Rightarrow B = -\frac{5}{2}$

$x = 2$ $10 = 12C \Rightarrow C = \frac{5}{6}$

$$I = \frac{5}{3} \int \frac{dx}{x+1} - \frac{5}{2} \int \frac{dx}{x+2} + \frac{5}{6} \int \frac{dx}{x-2}$$

$$= \frac{5}{3} \log|x+1| - \frac{5}{2} \log|x+2| + \frac{5}{6} \log|x-2| + C$$

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$^{**} 12] \int \frac{(x^3+x+1)dx}{x^2-1}$

$\circ \text{ of } nu > \circ \text{ of } de \Rightarrow IF$

$\begin{matrix} x^3+x+1 \\ x^2-1 \end{matrix} \xrightarrow{D} \begin{matrix} x^3+x \\ 2x+1 \end{matrix} \xrightarrow{R} \begin{matrix} x^3+x+1 \\ x^3+x \end{matrix} \xrightarrow{D} \begin{matrix} 1 \\ 2x+1 \end{matrix}$

$\frac{x^3+x+1}{x^2-1} = x + \frac{2x+1}{x^2-1} \rightarrow (1)$

$\frac{2x+1}{(x-1)(x+1)} = \frac{3}{2(x-1)} + \frac{-1}{2(x+1)}$

$(1) \Rightarrow I = \int x dx + \frac{3}{2} \int \frac{1}{x-1} dx + \frac{-1}{2} \int \frac{1}{x+1} dx$

$= \frac{x^2}{2} + \frac{3}{2} \log|x-1| + \frac{1}{2} \log|x+1| + C$

$13] \int \frac{2 \cdot dx}{(1-x)(1+x^2)}$

$\frac{2}{(1-x)(1+x^2)} = \frac{A}{1-x} + \frac{Bx+C}{1+x^2}$

$2 = A(1+x^2) + (Bx+C)(1-x)$

put $x=1: 2 = A(2) \Rightarrow A=1$

coeff of $x^2: 0 = A - B \Rightarrow B=1$

coeff of $x: 0 = B - C \Rightarrow C=1$

$I = \int \left(\frac{1}{1-x} + \frac{x+1}{1+x^2} \right) dx = \int \frac{1}{1-x} + \frac{x}{1+x^2} + \frac{1}{1+x^2} dx$

$= \int \frac{1}{1-x} + \frac{2x}{2(1+x^2)} + \frac{1}{1+x^2} dx$

$= \frac{\log|1-x|}{-1} + \frac{1}{2} \log|1+x^2| + \tan^{-1}x + C$

$$14) I = \int \frac{3x-1}{(x+2)^2} dx$$

$$\frac{3x-1}{(x+2)^2} = \frac{3(y-2)-1}{y^2} = \frac{3y-7}{y^2} \quad \text{put } x+2=y \\ x=y-2$$

$$= \frac{3}{y} - \frac{7}{y^2}$$

$$I = 3 \int \frac{1}{y} dy - 7 \int \frac{1}{y^2} dy$$

$$= 3 \log y + \frac{7}{y} + C$$

$$= 3 \log(x+2) + 7 \left(\frac{1}{x+2} \right) + C$$

$$15) \int \frac{1}{x^4-1} dx = \int \frac{1}{(x^2-1)(x^2+1)} dx = \int \frac{dx}{(x-1)(x+1)(x^2+1)}$$

$$\frac{1}{(x-1)(x+1)(x^2+1)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}$$

$$1 = A(x+1)(x^2+1) + B(x-1)(x^2+1) + (Cx+D)(x-1)(x+1)$$

$$\text{put } x=1: 1 = 4A \Rightarrow A = \frac{1}{4}$$

$$x=-1: 1 = -4B \Rightarrow B = -\frac{1}{4}$$

$$\text{coeff of } x^3: 0 = A+B+C \Rightarrow C=0$$

$$\text{coeff of } x^2: 0 = D \Rightarrow D=0$$

$$I = \frac{1}{4} \int \frac{dx}{x-1} - \frac{1}{4} \int \frac{dx}{x+1} = \frac{1}{4} \left(\log(x-1) - \log(x+1) \right) + C$$

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*** 16] $\int \frac{dx}{x(x^{n+1})}$ $\div$ & x by x^{n-1}

$$I = \int \frac{x^{n-1} dx}{x^n (x^{n+1})} \quad x^n = t$$

$$n x^{n-1} dx = dt$$

$$\frac{1}{n} \int \frac{dt}{t(t+1)} = \frac{1}{n} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt$$

$$I = \frac{1}{n} \left[\log t - \log(t+1) \right] + C$$

$$= \frac{1}{n} \log \left| \frac{t}{t+1} \right| + C$$

$$\boxed{\int \frac{dx}{x(x^{n+1})} = \frac{1}{n} \log \left| \frac{x^n}{x^{n+1}+1} \right| + C}$$

*] $\int \frac{dx}{x(x^7+1)}$ $\div$ & x by x^6

$$I = \int \frac{x^6 dx}{x^7 (x^7+1)} = \frac{1}{7} \int \frac{dt}{t(t+1)} \quad x^7 = t$$

$$7 x^6 dx = dt$$

$$\frac{1}{7} \int \left(\frac{1}{t} - \frac{1}{t+1} \right) dt = \frac{1}{7} \left[\log t - \log(t+1) \right] + C$$

$$= \frac{1}{7} \log \left| \frac{t}{t+1} \right| + C$$

$$\boxed{\int \frac{dx}{x(x^7+1)} = \frac{1}{7} \log \left| \frac{x^7}{x^7+1} \right| + C}$$

17] $\int \frac{\cos x dx}{(1-\sin x)(2-\sin x)}$

$$\sin x = t$$

$$\cos x dx = dt$$

$$I = \int \frac{dt}{(1-t)(2-t)}$$

$$\frac{1}{(1-t)(2-t)} = \frac{A}{1-t} + \frac{B}{2-t} = \frac{1}{1-t} - \frac{1}{2-t}$$

$$I = \int dt \left(\frac{1}{1-t} - \frac{1}{2-t} \right) = \log|1-t| - \log|2-t|$$

$$= \log \left| \frac{2-t}{1-t} \right| + C = \log \left| \frac{2-\sin x}{1-\sin x} \right| + C$$

$$18] I = \int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx$$

$$\frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} \quad \text{In partial fractions, } x^2 = t$$

$$\Rightarrow \frac{(t+1)(t+2)}{(t+3)(t+4)} = \frac{t^2+3t+2}{t^2+7t+12} \quad (\text{Improper fraction})$$

$$\frac{t^2+7t+12}{t^2+7t+12} - \frac{t^2+3t+2}{t^2+7t+12} = \frac{-4t-10}{t^2+7t+12} = -2(2t+5)$$

$$\frac{t^2+3t+2}{t^2+7t+12} = 1 - \frac{2(2t+5)}{(t+3)(t+4)} \rightarrow \textcircled{1}$$

$$\frac{2t+5}{(t+3)(t+4)} = \frac{A}{t+3} + \frac{B}{t+4} = \frac{-1}{t+3} + \frac{3}{t+4}$$

$$\therefore \int \frac{(x^2+1)(x^2+2)}{(x^2+3)(x^2+4)} dx = \int \left[1 - 2 \left(\frac{-1}{x^2+3} + \frac{3}{x^2+4} \right) \right] dx$$

$$= x + \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3}} \right) - \frac{6}{2} \tan^{-1} \left(\frac{x}{2} \right) + C$$

$$** 21] I = \int \frac{dx}{e^x - 1} = \int \frac{dt}{t(t-1)}$$

$$= \int \frac{-1}{t} + \frac{1}{t-1} dt$$

$$e^x = t$$

$$e^x dx = dt$$

$$dx = \frac{dt}{e^x} = \frac{dt}{t}$$

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$$= \log |t-1| - \log |t| + C$$

$$= \log \left| \frac{t-1}{t} \right| + C = \log \left| \frac{e^x-1}{e^x} \right| + C$$

19) $\int \frac{2x \, dx}{(x^2+1)(x^2+3)}$

$$\frac{2x}{(x^2+1)(x^2+3)} = \frac{(Ax+B)}{x^2+1} + \frac{(Cx+D)}{x^2+3}$$

$$2x = (Ax+B)(x^2+3) + (Cx+D)(x^2+1)$$

put $x^2 = t$

$$2x \, dx = dt$$

$$\int \frac{dt}{(t+1)(t+3)} \quad \text{Now do partial fractions.}$$

Bernoulli's rule:

$$\int uv dx = u \int v - u_1 \int \int v + u_2 \int \int \int v - \dots$$



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Eg 20] $\int \frac{x(\sin^{-1}x) dx}{\sqrt{1-x^2}}$

$$I = \int \frac{t \cdot \sin t dt}{\sqrt{1-x^2}}$$

$$= -t \cos t + \sin t + C$$

$$= -(\sin^{-1}x) \sqrt{1-x^2} + x + C$$

Definite Integrals

Fundamental theorem of integral calculus:

Theorem: Let f be a continuous function defined on $[a, b]$, and F be its antiderivative, then

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

EXERCISE 7.9

1] $I = \int_{-1}^1 (x+1) dx = \left[\frac{x^2}{2} + x \right]_{-1}^1 = \frac{1}{2}(1-1) + [1 - (-1)] = 2$

** 2] $I = \int_2^3 \frac{1}{x} dx = \log x \Big|_2^3 = \log 3 - \log 2 = \log \frac{3}{2}$

* $I = \int_1^2 \frac{1}{x} dx = \log x \Big|_1^2 = \log 2 - \log 1 = \log 2$

3] $I = \int_1^2 (4x^3 - 5x^2 + 6x + 9) dx = \left[\frac{4x^4}{4} - \frac{5x^3}{3} + \frac{6x^2}{2} + 9x \right]_1^2$

$$= (16-1) - \frac{5}{3}(8-1) + 3(4-1) + 9(2-1)$$

$$= 15 - \frac{35}{3} + 9 + 9 = 33 - \frac{35}{3} = \frac{64}{3}$$

4] $\int_0^{\frac{\pi}{4}} \sin 2x dx = \left[-\frac{\cos 2x}{2} \right]_0^{\frac{\pi}{4}} = -\frac{1}{2}(0-1) = \frac{1}{2}$

$$5) \int_0^{\frac{\pi}{2}} \cos 2x \cdot dx = \left[\frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}} = \frac{1}{2} [0 - 0] = 0$$

$$6) \int_4^5 e^x \cdot dx = e^x \Big|_4^5 = e^5 - e^4$$

$$7) \int_0^{\frac{\pi}{4}} \tan x \, dx = \log(\sec x) \Big|_0^{\frac{\pi}{4}} = \log \sqrt{2} - \log 1 = \log \sqrt{2} = \frac{1}{2} \log 2$$

$$8) \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \operatorname{cosec} x \cdot dx = \log(\operatorname{cosec} x - \cot x) \Big|_{\frac{\pi}{6}}^{\frac{\pi}{4}} = \log(\sqrt{2}-1) - \log(2-\sqrt{3}) = \log \left| \frac{\sqrt{2}-1}{2-\sqrt{3}} \right|$$

$$9) \int_0^1 \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x \Big|_0^1 = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

$$10) \int_0^1 \frac{dx}{1+x^2} = \tan^{-1} x \Big|_0^1 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$$

$$11) \int_2^3 \frac{dx}{x^2-1} = \frac{1}{2} \log \left| \frac{x-1}{x+1} \right| \Big|_2^3 = \frac{1}{2} \left[\log \left(\frac{2}{4} \right) - \log \left(\frac{1}{3} \right) \right] = \frac{1}{2} \log \left| \frac{1}{2} \cdot \frac{3}{1} \right| = \frac{1}{2} \log \left(\frac{3}{2} \right) = \log \sqrt{\frac{3}{2}}$$

$$12) \int_0^{\frac{\pi}{2}} \cos^2 x \cdot dx = \int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 2x}{2} \right) dx = \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}}$$

$$I = \frac{1}{2} \left\{ \left(\frac{\pi}{2} - 0 \right) + \frac{1}{2} (0 - 0) \right\} = \frac{\pi}{4}$$

$$13) I = \int_2^3 \frac{x \cdot dx}{x^2+1} = \frac{1}{2} \int_2^3 \frac{2x \cdot dx}{x^2+1} = \frac{1}{2} \log |x^2+1| \Big|_2^3 = \frac{1}{2} \{ \log 10 + \log 5 \} = \frac{1}{2} \log \left(\frac{10}{5} \right) = \frac{1}{2} \log 2 = \log \sqrt{2}$$

If a substitution is made for a definite integral, its limits also change.



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$$\begin{aligned}
 14) \int_0^1 \frac{2x+3}{5x^2+1} dx &= \int_0^1 \frac{2x}{5x^2+1} dx + \int_0^1 \frac{3}{5x^2+1} dx \\
 &= 2 \int_0^1 \frac{dt}{10(t^2+1)} + \frac{3}{5} \int_0^1 \frac{dx}{(x^2+\frac{1}{5})} \quad \begin{matrix} 5x^2+1=t \\ 10x \cdot dx=dt \\ x \cdot dx=\frac{dt}{10} \end{matrix} \\
 &= \frac{2}{10} \int_0^1 \frac{dt}{t^2+1} + \frac{3}{5} \int_0^1 \frac{dx}{x^2+(\frac{1}{\sqrt{5}})^2} \\
 &= \left[\frac{1}{5} \log(5x^2+1) \right]_0^1 + \left[\frac{3\sqrt{5}}{5} \tan^{-1}(x\sqrt{5}) \right]_0^1 \\
 &= \frac{1}{5} \log 6 - \log 1 + \frac{3}{\sqrt{5}} \tan^{-1} \sqrt{5}
 \end{aligned}$$

$$\begin{aligned}
 15) \int_0^1 x \cdot e^{x^2} dx &= \int_0^1 \frac{1}{2} dt \cdot e^t = \frac{1}{2} [e^t]_0^1 \\
 &= \frac{1}{2} (e^1 - e^0) = \frac{1}{2} (e - 1)
 \end{aligned}$$

$x^2 = t$
 $2x \cdot dx = dt$
 $x \cdot dx = \frac{dt}{2}$
 put $x=0, t=0$
 $x=1, t=1$

$$\begin{aligned}
 16) \int_1^2 \frac{5x^2}{x^2+4x+3} dx &= 5 \int_1^2 \frac{x^2}{x^2+4x+3} dx \\
 &= 5 \int_1^2 \left(1 - \frac{(4x+3)}{(x+1)(x+3)} \right) dx \\
 &= 5 \left[x - \frac{A}{x+1} - \frac{B}{x+3} \right]_1^2 \\
 &= 5 \left[x + \frac{1}{2(x+1)} - \frac{9}{2(x+3)} \right]_1^2 \\
 &= 5 \left\{ x + \frac{1}{2} \log(x+1) - \frac{9}{2} \log(x+3) \right\}_1^2
 \end{aligned}$$

$\frac{x^2}{x^2+4x+3} = \frac{x^2}{(x+1)(x+3)}$
 $\frac{x^2}{(x+1)(x+3)} = \frac{A}{x+1} + \frac{B}{x+3}$
 $x^2 = A(x+3) + B(x+1)$
 $x^2 = Ax + 3A + Bx + B$
 $x^2 = (A+B)x + (3A+B)$
 $A+B=0$
 $3A+B=1$
 $2A=1 \Rightarrow A=\frac{1}{2}$
 $B=-\frac{1}{2}$
 $B=\frac{-1}{2} = -\frac{1}{2}$

$$\begin{aligned}
 &= 5 \left\{ (2-1) + \frac{1}{2} (\log 3 - \log 2) - \frac{9}{2} (\log 5 - \log 4) \right\} \\
 &= 5 \left\{ 1 + \log \sqrt{\frac{3}{2}} - \frac{9}{2} \log \left(\frac{5}{4} \right) \right\} \\
 17) \quad I &= \int_0^{\frac{\pi}{4}} (2 \sec^2 x + x^3 + 2) dx \\
 &= 2 \tan x \left[\frac{x^4}{4} + 2x \right]_0^{\frac{\pi}{4}} = 2 \left(\tan \frac{\pi}{4} - \tan 0 \right) + \left(\frac{(\frac{\pi}{4})^4}{4} - 0 \right) + (2 \frac{\pi}{4} - 0) \\
 &= 2 + \frac{\pi^4}{1024} + \frac{\pi}{2} \\
 18) \quad \int_0^{\pi} \sin^2 \frac{x}{2} - \cos^2 \frac{x}{2} dx &= \int_0^{\pi} -\cos x \left(\frac{x}{2} \right) dx = \int_0^{\pi} -\cos x \\
 &= -\sin x \Big|_0^{\pi} = 0 \\
 20) \quad \int_0^1 x \cdot e^x + \sin \left(\frac{\pi x}{4} \right) dx &= \left[x \cdot e^x - e^x \right]_0^1 - \left[\cos \left(\frac{\pi x}{4} \right) \cdot \frac{4}{\pi} \right]_0^1 \\
 &= (e - 0) - (e - 1) - \frac{4}{\pi} \left(\frac{1}{\sqrt{2}} - 1 \right) \\
 &= e - e + 1 - \frac{4}{\pi \sqrt{2}} + \frac{4}{\pi} \left(\frac{1}{\sqrt{2}} - 1 \right) \\
 19) \quad \int_0^2 \frac{6x+3}{x^2+4} dx & \text{ split \& integrate} \\
 &= \int_0^2 \left(\frac{A}{x+2} + \frac{B}{x+2} \right) dx = \int_0^2 \left(\frac{P}{(x+2)^2} + \frac{Q}{(x+2)} \right) dx = \int_0^2 \left(\frac{P}{(x+2)^2} + \frac{Q}{(x+2)} \right) dx \\
 &= \left[\frac{P}{-(x+2)} + Q \ln(x+2) \right]_0^2 = \left[\frac{P}{-(x+2)} + Q \ln(x+2) \right]_0^2 \\
 &= \left\{ \frac{P}{-(x+2)} + Q \ln(x+2) \right\}_0^2
 \end{aligned}$$

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$$\int_{-5}^5 |x+2| dx$$

20) $\int_1^{\sqrt{3}} \frac{dx}{1+x^2} = [\tan^{-1} x]_1^{\sqrt{3}} = \frac{\pi}{3} - \frac{\pi}{4} = 60^\circ - 45^\circ = \frac{\pi}{12}$

22) $\int_0^{\frac{2}{3}} \frac{dx}{4+9x^2} = \int_0^{\frac{2}{3}} \frac{dx}{(2)^2 + (3x)^2} = \frac{1}{(3)^2} [\tan^{-1}(\frac{3x}{2})]_0^{\frac{2}{3}}$
 $= \frac{1}{9} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{36}$

EXERCISE 7.10

1) $\int_0^1 \frac{x}{x^2+1} dx$ $x^2+1=t$ $x=0, t=1$
 $2x \cdot dx = dt$ $x=1, t=2$
 $x dx = \frac{1}{2} dt$

$$= \int_1^2 \frac{1}{2} \cdot \frac{dt}{t} = \frac{1}{2} (\log 2 - \log 1) = \frac{1}{2} \log 2 = \log \sqrt{2}$$

3) $\int_0^1 \sin^{-1} \frac{2x}{1+x^2} dx$ $x = \tan \theta$ $x=0, \theta=0$
 $\theta = \tan^{-1} x$ $x=1, \theta = \frac{\pi}{4}$
 $dx = \sec^2 \theta d\theta$

$$I = \int_0^{\frac{\pi}{4}} \sin^{-1} \left(\frac{2 \tan \theta}{1 + \tan^2 \theta} \right) \sec^2 \theta d\theta$$

$$= \int_0^{\frac{\pi}{4}} \sin^{-1}(\sin 2\theta) \sec^2 \theta d\theta$$

by parts

$$I = 2 \left[\theta \cdot \tan \theta - \int \tan \theta d\theta \right]_0^{\frac{\pi}{4}} = 2 \left[\theta \tan \theta - \log(\sec \theta) \right]_0^{\frac{\pi}{4}}$$

$$= 2 \left[\left(\frac{\pi}{4} - 0 \right) - (\log \sqrt{2} - \log 1) \right] = 2 \left[\frac{\pi}{4} - \log \sqrt{2} \right] = \frac{\pi}{2} - \log 2$$

$$\begin{aligned}
 4] \quad I &= \int_0^2 x \sqrt{x+2} \, dx & \begin{array}{l} x+2=t \\ x=t-2 \\ dx=dt \end{array} & \begin{array}{l} x=0, t=2 \\ x=2, t=4 \end{array} \\
 I &= \int_2^4 (t-2) \sqrt{t} \, dt \\
 &= \int_2^4 (t^{\frac{3}{2}} - 2\sqrt{t}) \, dt = \left[\frac{2}{5} t^{\frac{5}{2}} - (2) \frac{2}{3} t^{\frac{3}{2}} \right]_2^4 \\
 &= \frac{2}{5} [t^{\frac{5}{2}}] - \frac{4}{3} [t^{\frac{3}{2}}]_2^4 = \frac{2}{5} (32 - 4\sqrt{2}) - \frac{4}{3} (8 - 2\sqrt{2}) \\
 &= \left(\frac{64}{5} - \frac{32}{3} \right) - \left(\frac{8}{5} + \frac{8}{3} \right) \sqrt{2} = \frac{32}{15} - \frac{8}{15} (8\sqrt{2}) = \frac{32-64\sqrt{2}}{15}
 \end{aligned}$$

compulsory
(6m) **

Properties of Definite Integrals

$$1] \quad P_1: \int_a^b f(x) \, dx = \int_a^b f(t) \, dt \quad (\text{change of variable does not affect the value of integral})$$

Proof:

$$LHS = \int_a^b f(x) \, dx = \int_a^b f(t) \cdot dt = RHS \quad \begin{array}{l} x=t \\ dx=dt \end{array} \quad \begin{array}{l} x=a, t=a \\ x=b, t=b \end{array}$$

$$2] \quad P_1: \int_a^b f(x) \, dx = - \int_b^a f(x) \, dx$$

$$\begin{aligned}
 LHS &= \int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a) = -[F(a) - F(b)] \\
 &= - \int_b^a f(x) \, dx = RHS
 \end{aligned}$$

$$3] \quad P_2: \int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx$$

$$\begin{aligned}
 LHS &= \int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a) \\
 &= F(b) - F(a) - F(c) + F(c) \\
 &= [F(b) - F(c)] + [F(c) - F(a)] \\
 &= \int_c^b f(x) \, dx + \int_a^c f(x) \, dx = RHS
 \end{aligned}$$

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4] $P_3: \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Eg: $I = \int_0^{\frac{\pi}{2}} \sin^2 x \cdot dx \xrightarrow{①} \int_0^{\frac{\pi}{2}} \sin^2(\frac{\pi}{2}-x) dx = \int_0^{\frac{\pi}{2}} \cos^2 x dx \rightarrow ②$

① + ②

$$2I = \int_0^{\frac{\pi}{2}} (\sin^2 x + \cos^2 x) dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$$\boxed{I = \frac{\pi}{4}} \quad \text{Hly } \int_0^{\frac{\pi}{2}} \cos^2 x \cdot dx = \frac{\pi}{4}$$

Proof: LHS = $\int_a^b f(x) dx$ put $x = a+b-t$ $\left| \begin{array}{l} x=a, t=b \\ x=b, t=a \end{array} \right.$
 $dx = -dt$

$$I = \int_b^a f(a+b-t) (-dt)$$

$$= - \int_b^a f(a+b-t) dt = \int_a^b f(a+b-t) dt \quad (\text{using } P_1)$$

$$= \int_a^b f(a+b-x) dx \quad (\text{using } P_0) = \text{RHS}$$

5] $P_4: \int_0^a f(x) dx = \int_0^a f(a-x) dx$

Proof: LHS = $\int_0^a f(x) dx$ put $x = a-t$ $\left| \begin{array}{l} x=0, t=a \\ x=a, t=0 \end{array} \right.$
 $dx = -dt$

$$I = \int_a^0 f(a-t) (-dt) = \int_0^a f(a-t) dt \quad (\text{using } P_1)$$

$$= \int_0^a f(a-x) dx \quad (\text{using } P_0) = \text{RHS}$$

$$6) P_5 : \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx$$

$$\text{LHS} = I = \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

$$\text{In 2nd integral : } x = 2a - t \quad \left| \begin{array}{l} x=a, t=a \\ x=2a, t=0 \end{array} \right.$$

$$dx = -dt$$

$$I = \int_0^a f(x) dx + \int_a^0 f(2a-t) (-dt)$$

$$= \int_0^a f(x) dx - \int_a^0 f(2a-t) dt = \int_0^a f(x) dx + \int_0^a f(2a-t) dt \text{ (using } P_1)$$

$$\therefore \int_0^{2a} f(x) dx + \int_0^a f(2a-x) dx \text{ (using } P_0)$$

$$(7) P_6 : \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

$$\text{Proof: LHS} = I = \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_a^{2a} f(x) dx$$

$$\text{In 2nd integral : } x = 2a - t \quad \left| \begin{array}{l} x=a, t=a \\ x=2a, t=0 \end{array} \right.$$

$$dx = -dt$$

$$I = \int_0^a f(x) dx + \int_a^0 f(2a-t) (-dt)$$

$$= \int_0^a f(x) dx - \int_a^0 f(2a-t) dt = \int_0^a f(x) dx + \int_0^a f(2a-t) dt \text{ (using } P_1)$$

$$\therefore \int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(2a-x) dx \text{ (using } P_0)$$

$$\text{case (i) } f(2a-x) = f(x)$$

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(x) dx = 2 \int_0^a f(x) dx$$

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EXERCISE 7.11

case (ii) If $f(2a-x) = -f(x)$

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx - \int_0^a f(x) dx = 0$$

cases (i) and (ii) $\Rightarrow$

$$\int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

8] P₁ : $\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & \text{if } f(x) \text{ is an even fun}^n \\ 0, & \text{if } f(x) \text{ is an odd fun}^n \end{cases}$

NOTE: Even funⁿ : If $f(-x) = f(x)$
Odd funⁿ : If $f(-x) = -f(x)$

LHS = $\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx$

In 1st integral : put $x = -t$ $\left| \begin{array}{l} x = -a, t = a \\ dx = -dt \\ x = 0, t = 0 \end{array} \right.$

$$I = \int_a^0 f(-t) (-dt) + \int_0^a f(x) dx$$

$$\int_{-a}^a f(x) dx = - \int_a^0 f(-t) dt + \int_0^a f(x) dx = \int_0^a f(-t) dt + \int_0^a f(x) dx \quad (P_1)$$

$$\int_{-a}^a f(x) dx = \int_0^a f(-x) dx + \int_0^a f(x) dx \quad (\text{using } P_0)$$

case (i) : If $f(x)$ is an even funⁿ, $f(-x) = f(x)$

$$\Rightarrow \int_{-a}^a f(x) dx = \int_0^a f(x) dx + \int_0^a f(x) dx = 2 \int_0^a f(x) dx$$

case (ii) : If $f(x)$ is an odd funⁿ, $f(-x) = -f(x)$

$$\int_{-a}^a f(x) dx = \int_0^a -f(x) dx + \int_0^a f(x) dx = 0$$

EXERCISE 1.11

Evaluate the following integrals.

$$1) \int_0^{\frac{\pi}{2}} \cos^2 x \cdot dx \xrightarrow{①} \int_0^{\frac{\pi}{2}} \cos^2\left(\frac{\pi}{2} - x\right) dx = \int_0^{\frac{\pi}{2}} \sin^2 x \cdot dx \rightarrow ②$$

① + ②

$$\mathcal{I} + \mathcal{I} = \int_0^{\frac{\pi}{2}} (\cos^2 x + \sin^2 x) dx$$

$$2\mathcal{I} = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$$\mathcal{I} = \frac{\pi}{4}$$

$$\text{Eg 3.4)} \mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sin^4 x}{\sin^4 x + \cos^4 x} dx \rightarrow ①$$

$$\mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sin^4\left(\frac{\pi}{2} - x\right) dx}{\sin^4\left(\frac{\pi}{2} - x\right) + \cos^4\left(\frac{\pi}{2} - x\right)} = \int_0^{\frac{\pi}{2}} \frac{\cos^4 x \cdot dx}{\cos^4 x + \sin^4 x} \rightarrow ②$$

① + ②

$$2\mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sin^4 x + \cos^4 x}{\sin^4 x + \cos^4 x} dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}}$$

$$2\mathcal{I} = \frac{\pi}{2} \Rightarrow \mathcal{I} = \frac{\pi}{4}$$

$$2) \mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} \cdot dx}{\sqrt{\sin x} + \sqrt{\cos x}} \rightarrow ①$$

$$\mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin\left(\frac{\pi}{2} - x\right)} dx}{\sqrt{\sin\left(\frac{\pi}{2} - x\right)} + \sqrt{\cos\left(\frac{\pi}{2} - x\right)}} = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\cos x} \cdot dx}{\sqrt{\cos x} + \sqrt{\sin x}} \rightarrow ②$$

① + ②

$$2\mathcal{I} = \int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}}$$

$$2\mathcal{I} = \frac{\pi}{2}$$

$$\mathcal{I} = \frac{\pi}{4}$$

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3)
$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x \, dx}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x} \rightarrow (1)$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}}(\frac{\pi}{2} - x) \, dx}{\sin^{\frac{3}{2}}(\frac{\pi}{2} - x) + \cos^{\frac{3}{2}}(\frac{\pi}{2} - x)} = \int_0^{\frac{\pi}{2}} \frac{\cos^{\frac{3}{2}} x \, dx}{\cos^{\frac{3}{2}} x + \sin^{\frac{3}{2}} x} \rightarrow (2)$$

(1) + (2)

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x} \, dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}}$$

$$2I = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

4)
$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^5 x \, dx}{\sin^5 x + \cos^5 x} \rightarrow (1)$$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos^5(\frac{\pi}{2} - x) \, dx}{\sin^5(\frac{\pi}{2} - x) + \cos^5(\frac{\pi}{2} - x)} = \int_0^{\frac{\pi}{2}} \frac{\sin^5 x \, dx}{\cos^5 x + \sin^5 x} \rightarrow (2)$$

(1) + (2)

$$2I = \int_0^{\frac{\pi}{2}} \frac{\sin^5 x + \cos^5 x}{\sin^5 x + \cos^5 x} \, dx = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}}$$

$$2I = \frac{\pi}{2} \Rightarrow I = \frac{\pi}{4}$$

7)
$$I = \int_0^1 x(1-x)^n \, dx$$

$$= \int_0^1 (1-x) [1-(1-x)]^n \, dx = \int_0^1 (1-x) x^n \, dx$$

$$= \int_0^1 x^n - x^{n+1} \, dx = \left[\frac{x^{n+1}}{n+1} - \frac{x^{n+2}}{n+2} \right]_0^1$$

$$= \frac{1}{n+1} - \frac{1}{n+2} = \frac{n+2 - n-1}{(n+1)(n+2)} = \frac{1}{(n+1)(n+2)}$$

$$\begin{aligned}
 \text{q) } I &= \int_0^2 x \sqrt{2-x} \, dx \\
 &= \int_0^2 (2-x) \sqrt{2-(2-x)} \, dx = \int_0^2 (2-x) \sqrt{x} \, dx \\
 &= \int_0^2 2\sqrt{x} - x^{3/2} \, dx = \left[2 \cdot \frac{2}{3} x^{3/2} - \frac{x^{5/2}}{5} \right]_0^2 \\
 &= \left[\frac{4}{3} x\sqrt{x} - \frac{2}{5} x^2\sqrt{x} \right]_0^2 = \frac{4}{3} (2\sqrt{2}) - \frac{2}{5} (4\sqrt{2}) \\
 &= 8\sqrt{2} \left(\frac{2}{15} \right) = \frac{16\sqrt{2}}{15}
 \end{aligned}$$

$$\begin{aligned}
 \text{*** 17) } I &= \int_0^a \frac{\sqrt{x} \cdot dx}{\sqrt{x} + \sqrt{a-x}} \rightarrow \textcircled{1} \\
 I &= \int_0^a \frac{\sqrt{a-x} \cdot dx}{\sqrt{a-x} + \sqrt{a-(a-x)}} = \int_0^a \frac{\sqrt{a-x} \cdot dx}{\sqrt{a-x} + \sqrt{x}} \rightarrow \textcircled{2} \\
 \textcircled{1} + \textcircled{2} \\
 2I &= \int_0^a \frac{\sqrt{a-x} + \sqrt{x}}{\sqrt{a-x} + \sqrt{x}} \, dx = \int_0^a 1 \, dx = [x]_0^a \\
 2I &= a \Rightarrow I = \frac{a}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{*) } \int_2^3 \frac{\sqrt{x} \cdot dx}{\sqrt{x} + \sqrt{5-x}} &\rightarrow \textcircled{1} \\
 I &= \int_2^3 \frac{\sqrt{5-x} \, dx}{\sqrt{5-x} + \sqrt{5-(5-x)}} = \int_2^3 \frac{\sqrt{5-x} \, dx}{\sqrt{5-x} + \sqrt{x}} \rightarrow \textcircled{2} \\
 \textcircled{1} + \textcircled{2} \\
 2I &= \int_2^3 \frac{\sqrt{5-x} + \sqrt{x}}{\sqrt{5-x} + \sqrt{x}} \, dx = \int_2^3 1 \, dx = [x]_2^3 \\
 2I &= 3-2 = 1 \\
 I &= \frac{1}{2}
 \end{aligned}$$

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15) $I = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx \rightarrow (1)$

$I = \int_0^{\frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x) - \cos(\frac{\pi}{2} - x)}{1 + \sin(\frac{\pi}{2} - x) \cos(\frac{\pi}{2} - x)} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \cos x \sin x} dx \rightarrow (2)$

$(1) + (2)$

$2I = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x + \cos x - \sin x}{1 + \sin x \cos x} dx = 0 \Rightarrow I = 0$

NOTE: $\int_a^b \frac{\sqrt{x} dx}{\sqrt{x} + \sqrt{b-x}} = \frac{b-a}{2}$

*** Eg 35) $I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan x}} \rightarrow (1)$

$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\tan(\frac{\pi}{2} - x)}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \sqrt{\cot x}} = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1 + \frac{1}{\sqrt{\tan x}}}$

$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\tan x} \cdot dx}{\sqrt{\tan x} + 1} \rightarrow (2)$

$(1) + (2)$

$2I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\tan x} + 1}{\sqrt{\tan x} + 1} dx = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 \cdot dx = x \Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}} = I$

$2I = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} \Rightarrow I = \frac{\pi}{12}$

*** 8) $I = \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx = \int_0^1 \frac{\log(1+x) dx}{1+x^2} = \frac{\pi}{8} \log 2$ (ET ***)

$I = \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$

$= \int_0^{\frac{\pi}{4}} \log\left[1 + \tan\left(\frac{\pi}{4} - x\right)\right] dx = \int_0^{\frac{\pi}{4}} \log\left[1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x}\right] dx$

$= \int_0^{\frac{\pi}{4}} \log\left[1 + \frac{1 - \tan x}{1 + \tan x}\right] dx = \int_0^{\frac{\pi}{4}} \log\left(\frac{1 + \tan x + 1 - \tan x}{1 + \tan x}\right) dx$

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{4}} \log\left(\frac{2}{1+\tan x}\right) dx = \int_0^{\frac{\pi}{4}} \log 2 \cdot dx - \int_0^{\frac{\pi}{4}} \log(1+\tan x) dx \\
 2I &= \log 2 \cdot \int_0^{\frac{\pi}{4}} 1 \cdot dx = \log 2 \left[x\right]_0^{\frac{\pi}{4}} = \log 2 \left(\frac{\pi}{4} - 0\right) \\
 2I &= \frac{\pi}{4} \log 2 \\
 I &= \frac{\pi}{8} \log 2
 \end{aligned}$$

~~1/8/17~~

5] $I = \int_{-5}^5 (x+2) dx$ if we put $x=-2$ then $(x+2)$ becomes zero
 $\therefore$ we put $x=-2.5$ and split the integration

$$\begin{aligned}
 &= \int_{-5}^{-2} -(x+2) dx + \int_{-2}^5 (x+2) dx \\
 &= \left[-\frac{x^2}{2} - 2x\right]_{-5}^{-2} + \left[\frac{x^2}{2} + 2x\right]_{-2}^5 \\
 &= -\frac{1}{2}(4-25) - 2(-2+5) + \frac{1}{2}(25-4) + 2(5+2) \\
 &= \frac{21}{2} + \frac{21}{2} - 6 + 14 = 21 + 8 = 29
 \end{aligned}$$

6] $I = \int_2^8 |x-5| dx = \int_2^5 -(x-5) dx + \int_5^8 (x-5) dx$

$$\begin{aligned}
 &= \left[-\frac{x^2}{2} + 5x\right]_2^5 + \left[\frac{x^2}{2} - 5x\right]_5^8 \\
 &= -\frac{1}{2}(25-4) + 5(3) + \frac{1}{2}(64-25) - 5(3) \\
 &= \frac{-21}{2} + \frac{39}{2} = \frac{18}{2} = 9
 \end{aligned}$$

18] $I = \int_0^4 |x-1| dx = \int_0^1 (1-x) dx + \int_1^4 (x-1) dx$

$$\begin{aligned}
 &= \left[x - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^4
 \end{aligned}$$

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$$= \frac{1}{2} + (8-4) - (-\frac{1}{2}) = \frac{1}{2} + 4 + \frac{1}{2} = 5$$

$$** 11] I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cdot dx \quad P_1: \int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx; & f \text{ is even} \\ 0 & ; f \text{ is odd} \end{cases}$$

$$f(-x) = [\sin(-x)]^2 = \sin^2 x = f(x)$$

$$\Rightarrow f \text{ is even} \rightarrow ①$$

$$I = 2 \int_0^{\frac{\pi}{2}} \sin^2 x \cdot dx = 2 \int_0^{\frac{\pi}{2}} \sin^2(\frac{\pi}{2} - x) dx$$

$$= 2 \int_0^{\frac{\pi}{2}} \cos^2 x \cdot dx \rightarrow ②$$

$$① + ②$$

$$2I = \int_0^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

$$I = \frac{\pi}{4}$$

$$** 13] I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \cdot dx$$

$$f(-x) = [\sin(-x)]^7 = -\sin^7 x = -f(x)$$

$$\Rightarrow f \text{ is an odd function.}$$

$$\Rightarrow I = 0$$

$$*** 20] I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^3 + x \cos x + \tan^5 x + 1) dx$$

$$I = 0 + 0 + 0 + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 \cdot dx = x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

$$Eg 33] I = \int_{-1}^1 (\sin^5 x \cdot \cos^4 x) dx$$

$$I = 0$$

$$14] \quad I = \int_0^{2\pi} \cos^5 x \cdot dx \quad P_6: \int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx & ; f(2a-x) = f(x) \\ 0 & ; f(2a-x) = -f(x) \end{cases}$$

$$f(2\pi-x) = [\cos(2\pi-x)]^5 = \cos^5 x = f(x)$$

$$\Rightarrow I = 2 \int_0^{\pi} \cos^5 x \cdot dx$$

$$f(\pi-x) = [\cos(\pi-x)]^5 = -\cos^5 x = -f(x) \text{ odd fun}^n$$

$$I = 0$$

$$\text{Eg 32]} \quad I = \int_0^{\pi} \frac{x \sin x \cdot dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x) dx}{1 + \cos^2(\pi-x)}$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin x \cdot dx}{1 + \cos^2 x} = \int_0^{\pi} \frac{\pi \sin x \cdot dx}{1 + \cos^2 x} - \int_0^{\pi} \frac{x \cdot \sin x \cdot dx}{1 + \cos^2 x}$$

$$2I = \pi \int_0^{\pi} \frac{\sin x \cdot dx}{1 + \cos^2 x} \quad \begin{array}{l} \cos x = t \\ -\sin x \cdot dx = dt \end{array} \quad \begin{array}{l} x=0, t=1 \\ x=\pi, t=-1 \end{array}$$

$$2I = \pi \int_1^{-1} \frac{-dt}{1+t^2} = -\pi [\tan^{-1} t]_1^{-1}$$

$$2I = -\pi \left(-\frac{\pi}{4} - \frac{\pi}{4} \right) = \pi \cdot \frac{\pi}{2}$$

$$I = \frac{\pi^2}{4}$$

$$\text{Eg 31]} \quad I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \cdot dx$$

$$f(x) = [\sin(x)]^2 = \sin^2 x = f(x)$$

$\Rightarrow f$ is even

$$I = 2 \int_0^{\frac{\pi}{4}} \sin^2 x \cdot dx = 2 \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos 2x}{2} \right) dx$$

$$= \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}} = \frac{\pi}{4} - \frac{1}{2}(1-0)$$

$$I = \frac{\pi}{4} - \frac{1}{2}$$

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$$\begin{aligned}
 12] \quad I &= \int_0^{\pi} \frac{x \, dx}{1 + \sin x} = \int_0^{\pi} \frac{(\pi - x) \, dx}{1 + \sin(\pi - x)} = \int_0^{\pi} \frac{(\pi - x) \, dx}{1 + \sin x} \\
 &= \int_0^{\pi} \frac{\pi \, dx}{1 + \sin x} - \int_0^{\pi} \frac{x \, dx}{1 + \sin x} \quad \text{--- (1)} \\
 2I &= \pi \int_0^{\pi} \frac{1}{1 + \sin x} \, dx \\
 2I &= \pi \int_0^{\pi} \left(\frac{1 + \sin x}{1 + \sin x} - \frac{\sin x}{1 + \sin x} \right) dx = \pi \int_0^{\pi} \frac{(1 - \sin x) \, dx}{\cos^2 x} \\
 &= \pi \int_0^{\pi} \sec^2 x - \sec x \cdot \tan x \, dx \\
 &= \pi \left[\tan x - \sec x \right]_0^{\pi} = \pi \left[(0 - 0) - (-1 - 1) \right] \\
 2I &= 2\pi \\
 I &= \pi
 \end{aligned}$$

** CET

Eg 36] $I = \int_0^{\frac{\pi}{2}} \log(\sin x) \, dx = \int_0^{\frac{\pi}{2}} \log(\cos x) \, dx = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log\left(\frac{1}{2}\right)$

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \log(\sin x) \, dx \quad \text{--- (1)} \\
 &= \int_0^{\frac{\pi}{2}} \log\left(\sin\left(\frac{\pi}{2} - x\right)\right) \, dx = \int_0^{\frac{\pi}{2}} \log(\cos x) \, dx \quad \text{--- (2)} \\
 \text{(1) + (2)} \\
 2I &= \int_0^{\frac{\pi}{2}} \log(\sin x) + \log(\cos x) \, dx = \int_0^{\frac{\pi}{2}} \log(\sin x \cdot \cos x) \, dx \\
 \{ \text{Let } 2I \} &= \int_0^{\frac{\pi}{2}} \log\left(\frac{\sin 2x}{2}\right) \, dx = \int_0^{\frac{\pi}{2}} \log(\sin 2x) \, dx - \int_0^{\frac{\pi}{2}} \log 2 \, dx \quad \text{--- (3)} \\
 I_1 &= \int_0^{\frac{\pi}{2}} \log(\sin 2x) \, dx \quad \begin{array}{l} 2x = t \\ 2 \cdot dx = dt \\ dx = \frac{1}{2} dt \end{array} \quad \begin{array}{l} x=0, t=0 \\ x=\frac{\pi}{2}, t=\pi \end{array}
 \end{aligned}$$

$$I_1 = \int_0^{\pi} \log(\sin t) dt \cdot \frac{1}{2} = \dots$$

$$f(\pi - t) = \log(\sin(\pi - t)) = \log(\sin t) = f(t)$$

$$\Rightarrow I_1 = 2 \int_0^{\frac{\pi}{2}} \log(\sin t) dt \cdot \frac{1}{2}$$

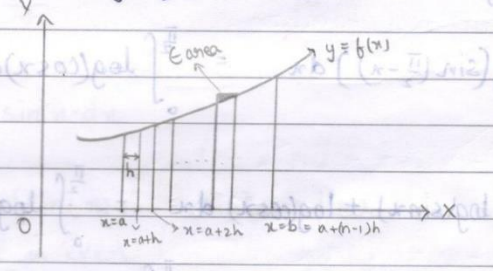
$$I_1 = \int_0^{\frac{\pi}{2}} \log(\sin x) dx = I_2 \text{ [using } P_0]$$

$$\textcircled{3} \Rightarrow 2I = I - \log 2 \int_0^{\frac{\pi}{2}} 1 \cdot dx$$

$$[(-1) - (0)] = -\log 2 [x]_0^{\frac{\pi}{2}} = -\log 2 \left(\frac{\pi}{2}\right)$$

$$\therefore \int_0^{\frac{\pi}{2}} \log(\sin x) dx = -\frac{\pi}{2} \log 2 = \frac{\pi}{2} \log \frac{1}{2}$$

Definite integrals as the limit of sum:
 The definite integral $\int_a^b f(x) dx$ is the area bounded by the curve $y = f(x)$, ordinates $x = a$, $x = b$ and x -axis.



$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \{ f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \}$$

where $h = \frac{b-a}{n} \Rightarrow nh = b-a$

$$0 = f, 0 = x$$

$$\pi = f, \frac{\pi}{2} = x$$

EXERCISE 7.8

Find the integral value as limit of sum

$$1) \int_a^b x \cdot dx ; f(x) = x, nh = b - a$$

$$\int_a^b f(x) dx = \lim_{h \rightarrow 0} h \{ f(a) + f(a+h) + f(a+2h) + \dots + f(a+(n-1)h) \}$$

$$\int_a^b x \cdot dx = \lim_{h \rightarrow 0} h \{ a + a+h + a+2h + \dots + a+(n-1)h \}$$

$$= \lim_{h \rightarrow 0} h \{ [a + a + a + \dots + n \text{ times}] + h(1+2+3+\dots+(n-1)) \}$$

AP: $a=1, d=1, n=n-1$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \lim_{h \rightarrow 0} h \left\{ na + h \frac{(n-1)(2+n-1)}{2} \right\}$$

$$= \lim_{h \rightarrow 0} \left\{ a(nh) + \frac{(nh-h)nh}{2} \right\}$$

$$= \lim_{h \rightarrow 0} \left\{ a(b-a) + \frac{(b-a-h)(b-a)}{2} \right\}$$

$$= \lim_{h \rightarrow 0} \left\{ ab - a^2 + \frac{(b-a)(b-a)}{2} \right\}$$

$$= \frac{2ab - 2a^2 + b^2 + a^2 - 2ab}{2} = \frac{b^2 - a^2}{2}$$

Verification

$$\left. \frac{x^2}{2} \right|_a^b = \frac{b^2 - a^2}{2}$$

$$2) I = \int_0^5 (x+1) dx ; f(x) = x+1, nh = 5-0 = 5 \frac{x^2}{2} + x \Big|_0^5 = \frac{25}{2} + 5 = \frac{35}{2}$$

$$\int_0^5 (x+1) dx = \lim_{h \rightarrow 0} h \{ f(0) + f(h) + f(2h) + \dots + f((n-1)h) \}$$

$$= \lim_{h \rightarrow 0} h \{ 1 + h + 1 + 2h + 1 + \dots + (n-1)h + 1 \}$$

$$= \lim_{h \rightarrow 0} h \{ (1+1+1+\dots+n \text{ times}) + h(1+2+3+\dots+(n-1)) \}$$

AP: $a=1, d=1, n=n-1$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \left\{ n + h \frac{(n-1)}{2} (2 + (n-1) - 1) \right\} \\
 &= \lim_{h \rightarrow 0} h \left\{ n + h \frac{(n-1)}{2} n \right\} = \lim_{h \rightarrow 0} \left\{ nh + nh \frac{(nh-h)}{2} \right\} \\
 &= \lim_{h \rightarrow 0} \left\{ 5 + 5 \frac{(5-h)}{2} \right\} = 5 + \frac{25}{2} = \frac{35}{2}
 \end{aligned}$$

$\star \star$ Eg 28] $\int_0^2 e^x \cdot dx$ $f(x) = e^x$, $nh = 2 - 0 = 2$

$$\begin{aligned}
 \int_0^2 e^x dx &= \lim_{h \rightarrow 0} h \left\{ f(0) + f(h) + f(2h) + \dots + f[(n-1)h] \right\} \\
 &= \lim_{h \rightarrow 0} h \left\{ 1 + e^h + e^{2h} + \dots + e^{(n-1)h} \right\} \\
 &\quad \text{GP: } a = e^h, r = e^h, S_n = a \frac{(r^n - 1)}{r - 1} \\
 &= \lim_{h \rightarrow 0} h \left\{ 1 + e^h \left(\frac{e^{n \cdot h} - 1}{e^h - 1} \right) \right\} \\
 &= \lim_{h \rightarrow 0} \left\{ h + h e^h \left(\frac{e^{nh} - 1}{(e^h - 1)} \right) \right\} \\
 &= \lim_{h \rightarrow 0} \left\{ h + e^h \left(\frac{e^{2-h} - 1}{(e^h - 1)} \right) \right\} \quad \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right) = 1 \\
 &= 0 + e^0 (e^2 - 1) = \boxed{e^2 - 1}
 \end{aligned}$$

5] $\int_{-1}^1 e^x \cdot dx$; $f(x) = e^x$, $nh = 1 - (-1) = 2$

$$\begin{aligned}
 \int_{-1}^1 e^x dx &= \lim_{h \rightarrow 0} h \left\{ f(-1) + f(-1+h) + f(-1+2h) + \dots + f[-1+(n-1)h] \right\} \\
 &= \lim_{h \rightarrow 0} h \left\{ e^{-1} + e^{-1+h} + e^{-1+2h} + \dots + e^{-1+(n-1)h} \right\} \\
 &= \lim_{h \rightarrow 0} h \left\{ e^{-1} + e^{-1} \cdot e^h + e^{-1} \cdot e^{2h} + \dots + e^{-1} \cdot e^{(n-1)h} \right\}
 \end{aligned}$$

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$$\begin{aligned}
 &= \lim_{h \rightarrow 0} h \{e^{-1} + e^{-1+h} + \dots + n \text{ times}\} \cdot xh(1+x)^{\frac{1}{x}-1} [2 \text{ eq}] \\
 &= \lim_{h \rightarrow 0} h \{e^{-1} [1 + e^h + e^{2h} + \dots + e^{(n-1)h}]\} \\
 &\quad \text{AP: } a=1, n=e^h, n=n-1, S_n = a \left(\frac{n^n - 1}{n-1} \right) \\
 &= \lim_{h \rightarrow 0} h \left\{ e^{-1} \cdot \frac{e^{nh} - 1}{e^h - 1} \right\} = \lim_{h \rightarrow 0} e^{-1} \left(\frac{e^2 - 1}{\frac{e^h - 1}{h}} \right) \\
 &= e^{-1} \left(\frac{e^2 - 1}{1} \right) = \frac{e^2 - 1}{e}
 \end{aligned}$$

3] $\int_2^3 x^2 dx$; $f(x) = x^2$, $nh = 3 - 2 = 1$. $\left[\frac{x^3}{3} \right]_2^3 = \frac{27}{3} - \frac{8}{3} = \frac{19}{3}$

$$\begin{aligned}
 \int_2^3 x^2 dx &= \lim_{h \rightarrow 0} h \{ f(2) + f(2+h) + f(2+2h) + \dots + f(2+(n-1)h) \} \\
 &= \lim_{h \rightarrow 0} h \{ 4 + (2+h)^2 + (2+2h)^2 + \dots + [2+(n-1)h]^2 \} \\
 &= \lim_{h \rightarrow 0} h \left\{ \underbrace{4+4+4h+h^2}_{\text{1st term}} + \underbrace{4+8h+4h^2}_{\text{2nd term}} + \dots + \underbrace{4+4(n-1)h+(n-1)^2h^2}_{\text{nth term}} \right\} \\
 &= \lim_{h \rightarrow 0} h \left\{ (4+4+\dots+n \text{ times}) + 4h[1+2+3+\dots+(n-1)] + h^2[1^2+2^2+\dots+(n-1)^2] \right\} \\
 &\quad \text{AP: } \sum n = \frac{n(n+1)}{2}, n=n-1 \quad \sum n^2 = \frac{n(n+1)(n+1)}{6} \\
 &= \lim_{h \rightarrow 0} h \left\{ 4n + 4h \frac{(n-1)n}{2} + h^2 \frac{(n-1)n[2(n-1)-1]}{6} \right\} \\
 &= \lim_{h \rightarrow 0} \left\{ 4nh + 2nh(nh-h) + \frac{nh(nh-h)(2nh-h)}{6} \right\} \\
 &= \lim_{h \rightarrow 0} \left\{ 4 + 2(1-h) + \frac{1(1-h)(2-h)}{6} \right\} \\
 &= 4 + 2(1) + \frac{1(1)(2)}{6} \\
 &= 6 + \frac{1}{3} = \frac{19}{3}
 \end{aligned}$$